Department of Physics, IIT Madras

PH1020 Physics II **Mid-term Examination** **23.03.2024**
Time: 9:30 am to 11:30 am ***Answer all questions*** **Max. marks: 30 Marks**

Name	Roll No.	Old Roll No.(if any)

Instructions: Each booklet should contain a 5 page question paper attached to an answer booklet containing six pages **(please check this now)**. The answers for **Part A** and **Part B** should be written in the allotted boxes in the question paper, **(for the objective questions we will not evaluate the answers if it is written in any place other than the allotted boxes)**. The answers to **Part C** should be written in the attached answer booklet. Vectors must be indicated clearly using arrows. You may use the reverse side of all pages in the question paper for rough work. All symbols have their usual meaning unless stated otherwise. You must use only black or blue ink for writing the answers and sketching the figures.Calculators, cell phones or any internet connectable device must not be in your possession during the examination.

Exam Hall No.

Total Marks

Useful Formulae

Cylindrical Polar coordinates:

$$\begin{aligned}
\vec{\nabla}u &= \frac{\partial u}{\partial \rho}\,\hat{e}_\rho + \frac{1}{\rho}\frac{\partial u}{\partial \varphi}\,\hat{e}_\varphi + \frac{\partial u}{\partial z}\,\hat{e}_z \\
\vec{\nabla}\cdot\vec{A} &= \frac{1}{\rho}\frac{\partial}{\partial \rho}(\rho A_\rho) + \frac{1}{\rho}\frac{\partial A_\varphi}{\partial \varphi} + \frac{\partial A_z}{\partial z} \\
\vec{\nabla}\times\vec{A} &= \left[\frac{1}{\rho}\frac{\partial A_z}{\partial \varphi} - \frac{\partial A_\varphi}{\partial z}\right]\hat{e}_\rho + \left[\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial \rho}\right]\hat{e}_\varphi + \frac{1}{\rho}\left[\frac{\partial}{\partial \rho}(\rho A_\varphi) - \frac{\partial A_\rho}{\partial \varphi}\right]\hat{e}_z
\end{aligned}$$

Spherical Polar coordinates:

$$\begin{aligned}
\vec{\nabla}u &= \frac{\partial u}{\partial r}\,\hat{e}_r + \frac{1}{r}\frac{\partial u}{\partial \theta}\,\hat{e}_\theta + \frac{1}{r\sin\theta}\frac{\partial u}{\partial \varphi}\,\hat{e}_\varphi \\
\vec{\nabla}\cdot\vec{A} &= \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 A_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(A_\theta \sin\theta) + \frac{1}{r\sin\theta}\frac{\partial A_\varphi}{\partial \varphi} \\
\vec{\nabla}\times\vec{A} &= \frac{1}{r\sin\theta}\left[\frac{\partial}{\partial \theta}(\sin\theta A_\varphi) - \frac{\partial A_\theta}{\partial \varphi}\right]\hat{e}_r \\
&\quad + \frac{1}{r}\left[\frac{1}{\sin\theta}\frac{\partial A_r}{\partial \varphi} - \frac{\partial}{\partial r}(r A_\varphi)\right]\hat{e}_\theta + \frac{1}{r}\left[\frac{\partial}{\partial r}(r A_\theta) - \frac{\partial A_r}{\partial \theta}\right]\hat{e}_\varphi
\end{aligned}$$

Part A

Clearly indicate your answers to each of the questions below in the boxes provided. The answers to each of these questions **must be a number or an expression.** Each box is worth 1 **mark.** [9 × 1 = 9 marks]

1. The screened Coulomb potential is given by

$$\Phi(r) = \frac{q}{4\pi\epsilon_0}\frac{\exp(-r/\lambda)}{r},$$

where r is the usual radial co-ordinate, and λ is a positive constant.

(a) The charge distribution $\rho(r)$ corresponding to this potential is:

$\rho(r) =$

(b) The total charge enclosed within a sphere of radius λ, with its origin at $r = 0$ is:

$Q_{\text{enc}} =$

2. A point dipole $\vec{\mathbf{p_1}} = p_1\hat{\mathbf{e}}_\mathbf{z}$ is held fixed at the origin O. Another point dipole $\vec{\mathbf{p_2}}$ with magnitude p_2 is located on the z-axis at a distance h above O, free to rotate in all directions. The minimum value of the potential energy for this configuration is:

$U_{\text{min}} =$

3. A square sheet of length L carrying a charge distribution $\sigma(x, y) = kx$, (with k a positive constant) is centred at the origin with its sides parallel to the Cartesian axis. The electric dipole moment of this configuration is:

$\vec{\mathbf{p}} =$

4. The electric polarization of a linear, isotropic, and homogenous dielectric material with electric susceptibility χ_e, relative permittivity $\epsilon_r = 3$ and electric displacement $\vec{\mathbf{D}} = 4 \times 10^{-7}\hat{\mathbf{e}}_\mathbf{x}$ C/m^2 is:

$\vec{\mathbf{P}} = \frac{8}{3} \times 10^{-7}\,\hat{e}_x$ C/m^2

$\bar{D} = \epsilon_0\bar{E} + \bar{P}$

$\bar{P} = \epsilon_0\chi_e\bar{E} \Rightarrow D = \epsilon_0\bar{E}(1+\chi_e)$

$D = \bar{P}(1 + 1/\chi_e)$

$\epsilon_r = 1 + \chi_e$

$\Rightarrow \chi_e = 2$

$P = \frac{D(\chi_e)}{\chi_e + 1}$

5. A sphere of radius R contains a dielectric material that carries a polarization given by

$$\vec{\mathbf{P}} = \frac{k}{r}\hat{\mathbf{e}}_r,$$

$\sigma_b = \vec{P}\cdot\hat{n} = k/\gamma$

$\rho_b = -\vec{\nabla}\cdot\vec{P} = -k/\gamma^2$

with k a positive constant.

(a) The bound volume charge density in the medium is given by:

$\rho_b =$ $-k/\gamma^2$

(b) The bound surface charge density at $r = R$ is given by:

$\sigma_b =$ k/γ

6. Given a charge distribution of the form

$$\rho(r,\theta,\varphi) = \begin{cases} \beta[1-(r^2/a^2)] & \text{if } 0 < r \le a \\ 0 & \text{if } a < r < \infty. \end{cases}$$

Here, β is a constant of appropriate dimensions

(a) The electric field in the region $0 < r \le a$ is

$\vec{\mathbf{E}} =$

(b) The electric field in the region $a < r < \infty$ is

$\vec{\mathbf{E}} =$

Part B

Each of the questions below are worth 2 marks. Pick the correct choice(s) for each of the questions below, and write your choices in the box provided for each question. Further, all constants have appropriate dimensions. Marks will be given only if no wrong choice/choices is/are picked. [$3 \times 2 = 6$ marks]

7. For a single point charge $+q$ placed at $\vec{r} = (1,0,0)$, which of the following statements are correct:

(A) The dipole moment is finite.
(B) The monopole moment is finite.
(C) The dipole moment is independent of the choice of origin.
(D) The quadrupole moment is zero.

2. For a conductor of arbitrary shape, pick the statements that hold true:
(A) The electrostatic potential inside a conductor should be strictly zero.
(B) The electric field on the surface of a conductor must be normal to the surface.
(C) The mobile charges distribute themselves on the surface because this is the configuration in which total electrostatic energy is minimum.
(D) If there exists a cavity of arbitrary shape inside the conductor, the net electric field in the conductor is non-zero.

3. It is known that the trajectory of a charged particle (starting from rest at the origin), moving under the combined effect of a crossed electric ($E\hat{e}_z$) and magnetic field ($B\hat{e}_x$) is described by a circle of radius R whose centre $(0, R\omega t, R)$ travels in the y direction, (with ω being a frequency). The following statements hold for R, and ω:

(A) $\omega \propto B$.
(B) $\omega \propto 1/B$.
(C) $R \propto 1/B^2$.
(D) $R \propto 1/B$.

Part C

These are questions that are subjective in nature. You need to show a clear work out of your solutions for this section in the answer sheets attached to the question paper. In the answer sheets clearly mark the question number that you are answering. All constants have appropriate dimensions. [15 marks]

1. Consider the following potential given by:

$$\Phi = \begin{cases} \frac{\rho_0}{4\pi\epsilon_0}\left(x^2+y^2+z^2\right) & \text{if } x^2+y^2+z^2 < a^2 \\ \frac{\rho_0}{4\pi\epsilon_0}\left(-a^2 + \frac{2a^3}{(x^2+y^2+z^2)^{1/2}}\right) & \text{if } x^2+y^2+z^2 > a^2 \end{cases}$$

Here, ρ_0 has the dimensions of a volume charge density and is a constant. Furthermore, ϵ_0 is the permittivity of free space. [3 marks]

(a) Calculate and plot the electric field corresponding to the above potential as a function of the radial co-ordinate r.

(b) Calculate the surface charge density at $r = a$.

2. An infinite slab of a linear isotropic homogenous dielectric material of relative permittivity ϵ_r, and thickness h is in contact with the surface of a conductor of infinite extent carrying a uniform charge density σ. Assume the top surface of the conductor is lying coincident with the xy plane. Find: [4 marks]

(a) The electric field both inside and outside the slab.

(b) The bound surface charge densities on both the top and bottom surface of the dielectric.

3. A wire centred at the origin is placed symmetrically along the z-axis. It carries a total charge Q, distributed via a linear charge density given by:

$$\lambda(z) = \begin{cases} \lambda_0 \left(1 - z^2/a^2\right) & \text{if } -a \leq z \leq a \\ 0 & \text{else.} \end{cases}$$

Here, the λ_0 is a positive constant of appropriate dimensions. Find the approximate potential upto quadrapolar order for the above charge distribution at the point $(0, 0, z)$, (assume $z \gg 2a$).

[5 marks]

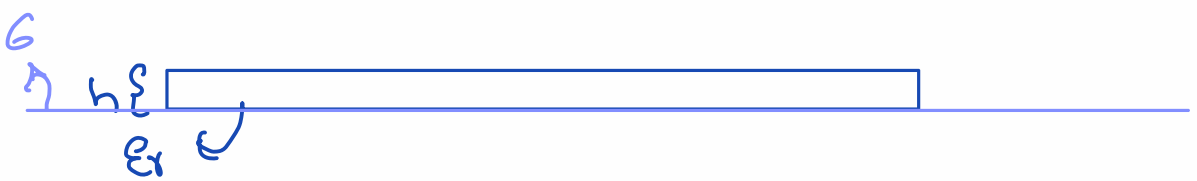
4. A system consists of a long solid inner cylinder of radius a, and a concentric very thin outer cylinder, of radius b. A uniform volume charge density ρ is distributed in the inner cylinder and outer cylinder has an equal but opposite charge pasted on the outer cylinder. Find the electrostatic energy per unit length in the system of cylinders.

[3 marks]

2. An infinite slab of a linear isotropic homogenous dielectric material of relative permittivity ϵ_r, and thickness h is in contact with the surface of a conductor of infinite extent carrying a uniform charge density σ. Assume the top surface of the conductor is lying coincident with the xy plane. Find: [4 marks]

(a) The electric field both inside and outside the slab.

(b) The bound surface charge densities on both the top and bottom surface of the dielectric.



(a) $E_{outside} = \frac{\sigma}{\epsilon_0}$

$E_{inside} = \frac{\sigma}{\epsilon_r \epsilon_0}$

(b) $\sigma_b = \vec{P}\hat{n}$

$\vec{P} = \epsilon_0 \chi_e \vec{E}$

$= \epsilon_0(\epsilon_r - 1)\vec{E}$

top: $\sigma(1 - 1/\epsilon_r)$

bottom: $-\sigma(1 - 1/\epsilon_r)$

Q5 has not been corrected
It is written before Q4

Department of Physics, IIT Madras

PH1020 (Physics II) **Quiz I** **20/04/2023**

Time: 8:00 – 8:50 am **Answer All questions** **Total Marks: 20**

Name	Roll. No.	Exam Hall. No.
Aryan Muralidharan	ME22B109	RMN 102

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Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Total marks
2	2	2	2	NA +2	0	2	3	2	15+2 = sid.

Instructions:

(i) **Vectors must be indicated clearly using arrows.**

(ii) You must use only **black or blue ink** for writing the answers.

(iii) You are NOT ALLOWED to carry calculators, cell phones and other internet enabled devices.

(iv) Necessary information on the curvilinear coordinate systems as well as important vector derivatives and identities are listed on the last page.

For the questions Q1 – Q3, write the answers in the allotted box.

Q1: Write True or False. **[2 × 1 = 2 *marks*]**

(I) If the total charge is zero in a local distribution, the dipole moment of this distribution varies with the choice of the origin.

False

(II) $\vec{E} = ky\hat{x} - kx\hat{y}$, where k is a constant of appropriate dimension, is a valid electrostatic field.

False

$= \frac{Q_{enc}}{\epsilon_0}$

1

Rough

$\nabla \cdot \vec{E} = 0$

Rough

$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial_x & \partial_y & \partial_z \\ ky & -kx & 0 \end{vmatrix}$ $\hat{i}(-k-k) = -2k$

[2 × 1 = 2 marks]

Q2: Fill in the Blanks

(I) The inward and outward flux for a closed surface are measured to be ϕ_1 and ϕ_2 respectively. The net charge enclosed is

$\varepsilon_0(\phi_2 - \phi_1)$

(II) For a charged spherical conductor of radius R, with an irregularly shaped cavity containing a charge q, the surface charge density on the outer surface is

$\frac{+q}{4\pi R^2}$

[2 × 1 = 2 marks]

Q3: Multiple Choice Questions

No marks will be awarded if any of the chosen option(s) is incorrect.

(I) Consider a spherical homogeneous linear dielectric with a point charge $+q$ at its center. The radius of the sphere is a and its permittivity is ϵ. Which of the following statements are correct?

(A) $\rho_b = 0$ and $\sigma_b = -\frac{q}{4\pi a^2}$

(B) ρ_b is negative and $\vec{D}$ is directed radially outwards.

(C) $\vec{\nabla} \cdot \vec{E} = \frac{\rho_b}{\epsilon_0(1+\chi_e)}$ for $0 < r < a$. χ_e is the electric susceptibility.

(D) $\vec{\nabla} \times \vec{D} = 0$ and $\vec{E} = \frac{q}{4\pi\epsilon r^2}\hat{r}$, for $0 < r < a$

B, D

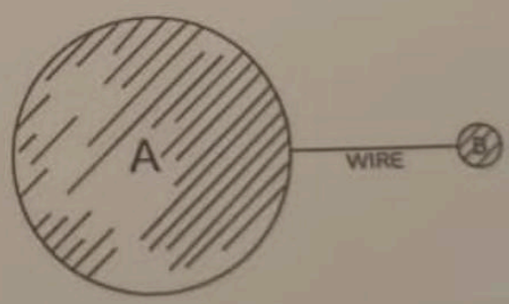
(II) There are two charged conducting spheres A and B having unequal radii and joined by a thin conducting wire as shown in the figure below. If V_A and V_B are the potentials at the surface of A and B, respectively and σ_A and σ_B are the corresponding surface charge densities, which of the following statements are true?

(A) $V_A = V_B, \sigma_A = \sigma_B$

(B) $V_A = V_B, \sigma_A < \sigma_B$

(C) $V_A \neq V_B, \sigma_A > \sigma_B$

(D) $V_A \neq V_B, \sigma_A = \sigma_B$



B

For the questions Q4 – Q9, write the answers in the attached booklet. Write your Name and Roll. No. on the first page of the booklet.

Q4 [2 marks]: Plot the potential (V) and electric field (E) as a function of the radial distance due to a solid metal sphere of unit radius and unit charge in all space. Write the values of E and V at the boundary of the sphere.

Q5 [2 marks]: A metal sphere of radius R carries a total charge Q. Calculate the force of repulsion between the northern and the southern hemispheres.

Q6 [2 marks]: Consider a homogeneous linear dielectric with permittivity ϵ. Let A be a point at the surface boundary between the dielectric and vacuum. The electric field $\vec{E}_0$ in the vacuum makes an angle α with the normal to the surface at the point A. Calculate the intensity of the electric field very close to the point A inside the dielectric.

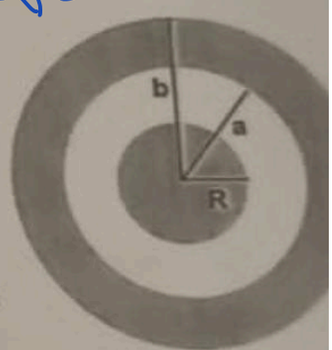
$D^{\perp}_{above} = D^{\perp}_{below}$ $E^{\parallel}_{above} = E^{\parallel}_{below}$

$\varepsilon_0 E_0 \cos\theta = \varepsilon E_{below}$ $E_0 \sin\theta = E_{below}$

$E = (E_\perp + E_\parallel) = E_0 \sin\alpha + \left(\frac{\varepsilon_0 E_0}{\varepsilon}\right)^2 \cos^2\theta = \frac{E_0}{\varepsilon}\sqrt{\varepsilon^2 \sin^2\alpha + \varepsilon_0^2 \cos^2\theta}$

Q7 [2 marks]: Consider a solid dielectric right-circular cone with half angle α and height h. The cone has uniform polarization P parallel to its axis of symmetry and pointing towards its base. Calculate the total bound charge Q_b on the curved surface of the cone.

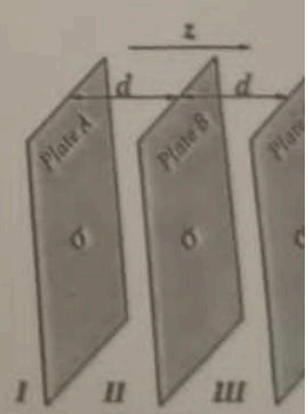
Q8 [3 marks]: A metal sphere, carrying charge q, is surrounded by a thick concentric metal shell as shown in the figure on the right. The shell carries no net charge.



(a) Calculate the potential at the center considering the reference point at infinity.

(b) If the outer shell is grounded, what will be the new potential at the center?

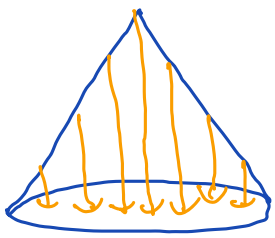
Q9 [3 marks]: Three large parallel plates, A, B, and C, with identical surface charge density σ are placed, with adjacent plates separated by a distance d, as shown in the figure on the right.



(a) Calculate the electric field in the regions *I*, *II*, *III*, and *IV*.

(b) What is the force per unit area on each plate?

Q7 [2 marks]: Consider a solid dielectric right-circular cone with half-angle α and height h. The cone has uniform polarization $\vec{P}$ parallel to its axis of symmetry and pointing towards its base. Calculate the total bound charge Q_b on the curved surface of the cone.



$\rho_b = 0$ (uniform)

σ_b (base) $= \vec{P}\cdot\hat{n} = P$

$q_{net} = 0$

$\Rightarrow \int \rho_b \, d\tau + \int \sigma_b \, da = 0$ (the $\int \rho_b \, d\tau$ term $\to 0$)

$\Rightarrow P \times \pi h^2 \tan^2\alpha + \int \sigma_b(\text{curved}) \, da = 0$

$\Rightarrow q_{\text{curved bounded}} = -P\pi h^2 \tan^2\alpha$

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A

Roll No. | M | E | 2 | 2 | 8 | 1 | 0 | 9 |

Name : Aryan Muralidharan

Quiz I [✓] Quiz II/ Mid-Sem [ ] End-Semester [ ] Make-up [ ] Total No. of Pages [ ]

Semester & Degree : [ ] Course No. [ ] Date : [ ] Part : [ ]

Question No.	1	2	3	4	5	6	7	8	9	10
Marks										

11	12	13	14	15	16	17	18	19	20	Total

Answer on both sides of the paper including the space below

Q4

R = 1 unit

Q = 1 unit

$\vec{E} = 0, \quad 0 \le r < 1$

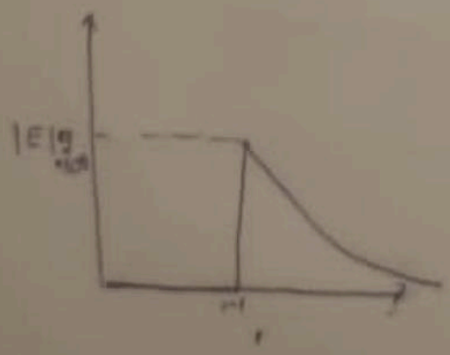
$= \frac{9\times10^9}{r^2} = \frac{K}{r^2}\hat{r}, \quad r \ge 1$

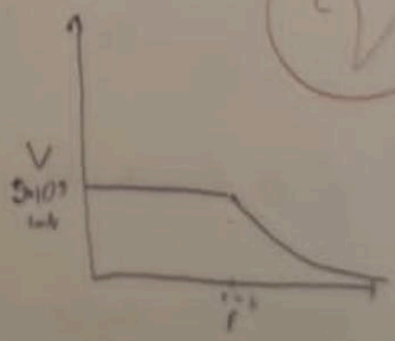
$V = 9\times10^9 \text{ units} = K \text{ units}, \quad 0 \le r < 1$

$\frac{9\times10^9}{r} = \frac{K}{r} \text{ units}, \quad r \ge 1$

At boundary, $\vec{E} = (9\times10^9 \text{ units})\,\hat{r}$ (radially outwards)

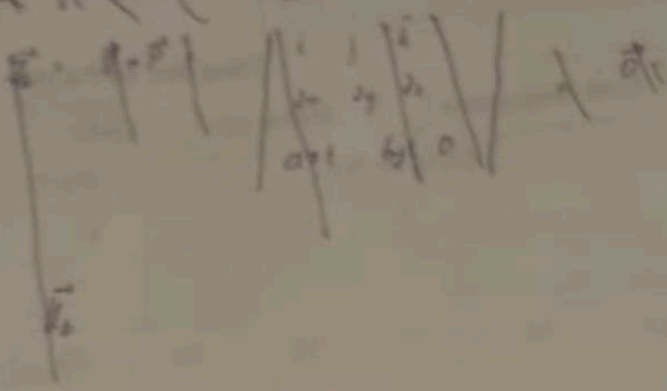
$V = (9\times10^9 \text{ units})$





(2)

4. $\vec{P} = ax\hat{i} + by\hat{j}$



5. $\vec{P} = ax\hat{i} + by\hat{j}$

$\rho_b = -\vec{\nabla}\cdot\vec{P} = -(a+b)$

$\sigma_b = \vec{P}\cdot\hat{n} = (ax\hat{i} + by\hat{j})\cdot\hat{z} = 0$

$\rho_0 = -(at + bt) = -(a+b)t$

By applying equation of continuity:

$$\vec{\nabla}\cdot\vec{J} = -\frac{\partial\rho}{\partial t}$$

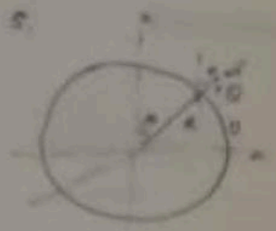
2

Here $\vec{\nabla}\cdot\vec{J}_b = (a+b)$ units $-①$

$$-\frac{\partial\rho}{\partial t} = -\frac{\partial}{\partial t}-(a+b)t = (a+b) \quad -②$$

From ① & ②: eqn of cont is satisfy $\Rightarrow$ bound charge is conserved.

6.



Electric field at point = $\frac{\sigma^2}{2\varepsilon_0}\hat{r}$ ✓

$$\int d\vec{F} = \int_0^{2\pi}\int_0^{\pi/2} \frac{\sigma^2}{2\varepsilon_0}\cos\theta\, R^2 \sin\theta\, d\theta\, d\phi\, (\hat{z})$$

$$\int_0^{\pi/2} \sin\theta\cos\theta\, d\theta = \frac{1}{2}(-\cos 2\theta)\Big|_0^{\pi/2}$$

$$\frac{1}{2}[1] = \frac{1}{2}$$

$$\vec{F} = \frac{\sigma^2 R^2}{2\cdot 2\varepsilon_0} \times 2\pi \int_0^{\pi/2} \sin 2\theta\, d\theta\, \hat{z}$$ ✓

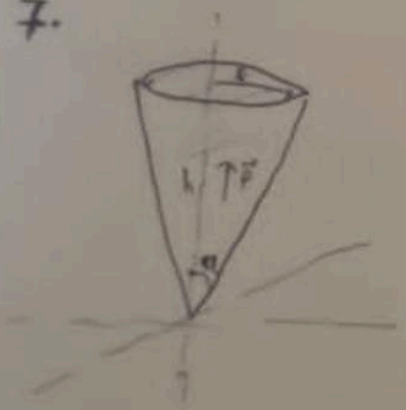
$$\vec{F} = \frac{\sigma^2 R^2 \pi}{2\varepsilon_0}\hat{z}$$

(2)

$$\sigma = \frac{Q}{4\pi R^2} \quad \therefore \quad \vec{F} = \frac{Q^2}{16\pi^2 R^2}\cdot\frac{R^2\pi}{2\varepsilon_0}\cdot 2 = \frac{Q^2}{64\varepsilon_0\pi R^2}\cdot 2\,(\hat{z})$$

$$\vec{F} = \frac{Q^2}{32\pi R^2\varepsilon_0}\hat{z}$$ ✓

7.



Here $\vec{P}$ is uniform

$$\therefore \rho_b = -\vec{\nabla}\cdot\vec{P} = 0$$ ✓

On the base of cone

$$\sigma_b = \vec{P}\cdot\hat{n} = P_{unif}$$ ✓

Also, $\tan\alpha = \frac{R}{h} \Rightarrow R = h\tan\alpha$.

for cone $\int \rho_b dV = \int \sigma_b dA = 0$

$$\Rightarrow \int_{base} \sigma_b\, dA + \int_{curved} \sigma_b\, dA = 0$$ ✓

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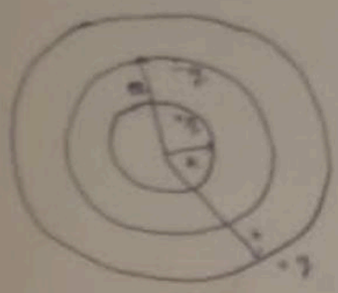
Indian Institute of Technology Madras

17.5
+0.5
= 18

$\therefore$ Q enclosed surface $= -(\rho_0 \pi (b\sin\alpha)^2)$

(2) $\quad$ Q shell $= -\rho(\pi R^2 \sin^2\alpha)$

1



By Gauss Law

(a) $\vec{E} = \begin{cases} 0, & r \le a \\ \frac{kq}{r^2}\hat{r}, & a \le r \le b \\ 0, & a \le r \le b \\ \frac{kq}{r^2}\hat{r}, & r \ge b \end{cases}$

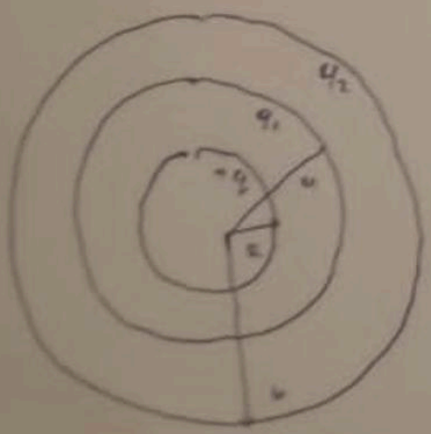
$V(\text{inside}) = -\int_{\infty}^{r} \vec{E}\cdot d\vec{r} = -\int_{\infty}^{b} \frac{kq}{r^2} dr - \int_{a}^{r} \frac{kq}{r^2} dr$

$$\boxed{V(\text{inside}) = \frac{kq}{b} + \frac{kq}{R} - \frac{kq}{a}}$$

(b) If outer shell is grounded, $V(\text{shell}) = 0$

Let new charges at the inner surface $= q_1$

outer surface $= q_2$



$V(\text{outside shell}) = 0$

$$\therefore \frac{kq}{b} + \frac{kq_1}{b} - \frac{kq_2}{b} = 0$$

$$q_1 + q_2 = -q \quad -(1)$$

$V(\text{inside shell}) = 0$

$$\frac{kq}{a} + \frac{kq_1}{a} + \frac{kq_2}{b} = 0$$

$$q + q_1 = -\frac{a}{b} q_2$$

Solving both we get:

$$\boxed{\begin{aligned} q_1 &= -q \\ q_2 &= 0 \end{aligned}}$$

25/05/2023

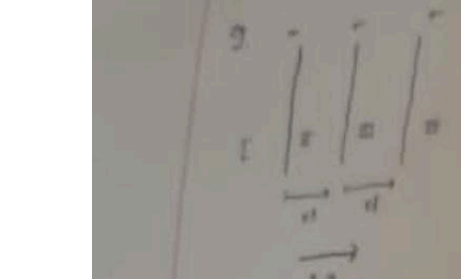
Marks: 20

xam Hall. No.

$V(\text{new})$ at centre $= \frac{kq}{R} - \frac{kq}{a}$ ✓3

$V = kq\left\{\frac{1}{R} - \frac{1}{a}\right\}$

9. (a)



By Super-position principle:

$(\vec{E})_{\text{net at pt}} = (\vec{E})_1 + (\vec{E})_2 + (\vec{E})_3$

At region I:

$\vec{E} = \frac{\sigma}{2\epsilon_0}(-\hat{z}) + \frac{\sigma}{2\epsilon_0}(-\hat{z}) + \frac{\sigma}{2\epsilon_0}(-\hat{z})$

$\therefore \vec{E}_I = \frac{3\sigma}{2\epsilon_0}(-\hat{z})$ ✓

At II:

$\vec{E} = \frac{\sigma}{2\epsilon_0}(+\hat{z}) - 2\times\frac{\sigma}{2\epsilon_0}\hat{z}$

$\therefore \vec{E}_{II} = -\frac{\sigma}{2\epsilon_0}\hat{z}$ ✓

(2)

At III:

$\vec{E} = \frac{\sigma}{2\epsilon_0}\hat{z} + \frac{\sigma}{2\epsilon_0}\hat{z} - \frac{\sigma}{2\epsilon_0}\hat{z}$

$\therefore \vec{E}_{III} = \frac{\sigma}{2\epsilon_0}\hat{z}$ ✓

At IV:

$\vec{E}_{IV} = +\frac{3\sigma}{2\epsilon_0}\hat{z}$ ✓

(b) Force per unit area = electrostatic pressure =

$\boxed{\vec{f} = \frac{\sigma^2}{2\epsilon_0}(+\hat{z})}$ ✗

This is same for each plate as they have same value of σ

Department of Physics, IIT Madras

25/05/2023

Quiz II

PH1020 (Physics II) **Total Marks: 20**

Time: 8:00 – 8:50 am Answer All Questions

Name	Roll No. *[If you have changed branch, write both old and new Roll No.]*	Exam Hall. No.
Vansh Jain	Old → ME22B136 New – EE22B164	RMN 201

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Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Total marks
2	3	2	2	0.5	3	3	3	~~18~~ 18.5

Instructions:

(i) **Vectors must be indicated clearly using arrows.**

(ii) You must use only **black or blue ink** for writing the answers. USE OF PENCIL IS NOT ALLOWED to write the answers and sketch the plots.

(iii) You are NOT ALLOWED to carry calculators, cell phones and other internet enabled devices.

(iv) Necessary information on the curvilinear coordinate systems, important vector derivatives and identities, as well as **some important formulas** are provided on the last page for your use.

(v) All symbols have their usual meaning unless stated otherwise. All constants are of appropriate dimensions. Further assume that unless otherwise stated, all constants are positive.

For the questions Q1 – Q3, write the answers in the allotted box.

[$2 \times 1 = 2$ *marks*]

Q1. Write **True** or **False.**

(I) A charged particle experiences an acceleration when it moves in a direction normal to the magnetic field.

True ✓ 1

(II) In matter, $\vec{\nabla} \times \vec{H} = \vec{J}_f + \epsilon_0 \frac{\partial \vec{E}}{\partial t}$

False ✓ 1

[$3 \times 1 = 3$ *marks*]

Q2. Fill in the blanks

(I) A long cylindrical conductor carries a volume current density $J = \frac{J_0}{s}$ along the axis of the cylinder. Here J_0 is a constant and s is the radial distance. The auxiliary field $\vec{H}$ within the conductor is:

$J_0 \hat{\phi}$ ✓ 1

Hint : $\int \frac{J_0}{s} \times s\,d\rho\,d\phi$ $\quad H = J_0$

(18)

(ii) A very long solenoid carrying current I is filled with a linear magnetic material of susceptibility $\chi_m = -1$. The magnetic fields inside and outside the solenoid, respectively, are:

Outside B = 0 ✓ Inside : B = 0 ✓

(iii) A material is kept in a time varying electric field $E = E_0 \cos 5t$, where E_0 is a constant and there is no external magnetic field. How much should be the volume current density J, supplied through the material so that it does not produce any magnetic field? Take ϵ_r and μ_r to be 1.

$J = 5\epsilon_0 E_0 \sin 5t$ ✓

Q3. Multiple Choice Questions [*No marks will be awarded if any of the chosen option(s) is incorrect.*] [$2 \times 1 = 2$ *marks*]

(i) Which of the following statements about the magnetic vector potential $\vec{A}$ are always true?

(A) The magnetic vector potential is uniquely defined for a given magnetic field. infinitely

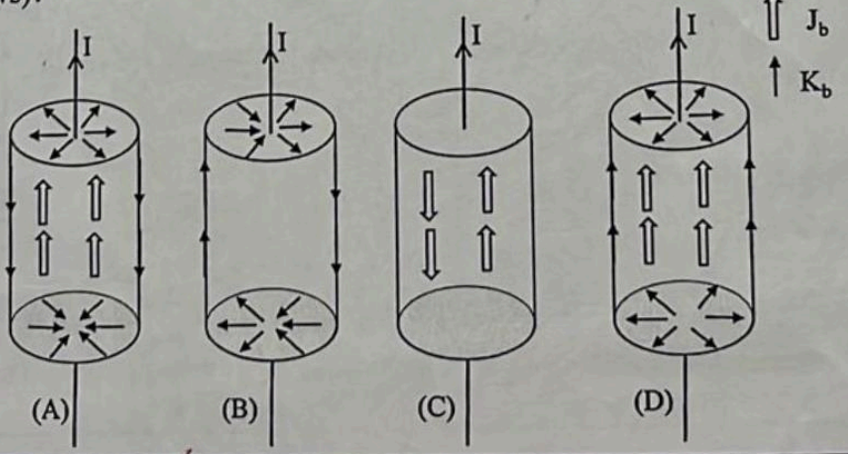
(B) Tangential components of $\vec{A}$ are always continuous across a boundary.

(C) $\oint_L \vec{A}.\vec{dl}$ equals the magnetic flux across a surface bounded by the loop of length L.

(D) The direction of the vector potential is normal to the direction of the current flow. along

(B) (C) ✓

(ii) An infinitely long wire carrying a current I passes through the axis of a cylinder of finite size. The cylinder is made out of a linear paramagnetic material. Which of the following sketches correctly represent the flow of volume bound current $\vec{J}_b$ (open arrows) and surface bound current $\vec{K}_b$ (solid arrows)?



(A) ✓

For the questions Q4 – Q8, write the answers in the attached bookiet. Write your Name and Roll. No. on the first page of the bookiet.

Q4. [2 marks] The magnetic field inside a long solid cylindrical conductor of radius R is found to be, $\vec{B} = B_0\hat{\phi}$, where B_0 is a constant. Determine the radial dependence of the volume current density $\vec{J}$ and plot it.

$\partial\rho_b/\partial t + \vec{\nabla}\cdot\vec{J}_b = 0$ $\quad J_b = dP/dt = a\hat{x}\hat{x} + a\hat{y}\hat{y}$

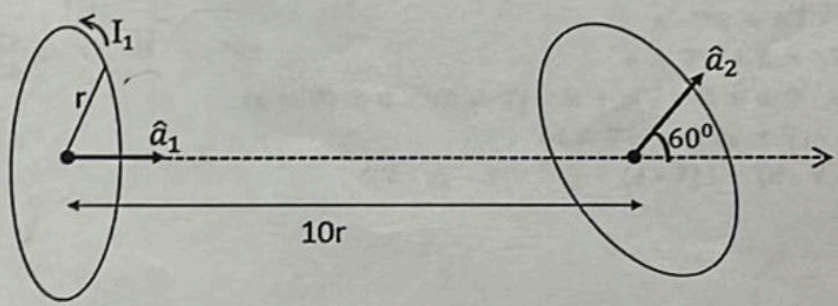
Q5. [2 marks] The time-varying polarization in a given material is found to be $\vec{P} = axt\,\hat{x} + byt\,\hat{y}$. Here, a and b are constants of appropriate dimensions. Show that current produced by $\vec{P}$ conserves the bound charges.

$\rho_b = -\vec{\nabla}\cdot\vec{P}$

Q6. [3 marks] A thin disk of radius a, with a uniform surface charge density σ, is rotating about its axis (z-axis) with a constant angular frequency ω. There is a uniform magnetic field $\vec{B} = B_0\hat{y}$ perpendicular to the axis of rotation. Determine the torque on the disk.

Q7. [3 marks]. Two electrons, each possessing a magnetic dipole moment $\vec{m} = m\hat{z}$, are constrained to move along the z-axis. In the absence of any other force, after reaching the equilibrium, how far away will they be from each other?

Q8. [3 marks] Consider two **tiny circular wire loops,** each of radius 'r' and with their center lying on the x-axis as shown in the figure below. The loops are separated by a distance 10r. $\vec{a}_1$ and $\vec{a}_2$ are the area vectors of loops 1 and 2, respectively. Let I_1 be the current flowing in loop1.



(a) Find the mutual inductance.

(b) Now if a current I_2 is turned-on in loop-2, how much work must be done to keep the current I_1 flowing in loop-1?

eE $\quad E\cdot dl =$ $\quad \oint$

$\frac{kqq_1}{r^2}$

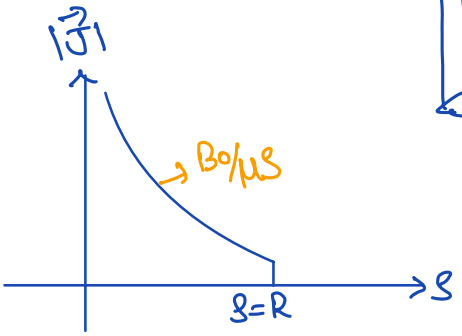
Q4. [2 marks] The magnetic field inside a long solid cylindrical conductor of radius R is found to be, $\vec{B} = B_0\hat{\phi}$, where B_0 is a constant. Determine the radial dependence of the volume current density $\vec{J}$ and plot it.

$$\vec{\nabla}\times\vec{B} = \mu_0\vec{J}$$

$$\frac{1}{s}\frac{\partial}{\partial s}(sB_0)\hat{e_z} = \mu_0\vec{J}$$

$$\Rightarrow \frac{B_0}{s}\hat{z} = \mu_0\vec{J}$$

$$\Rightarrow \vec{J} = \frac{B_0}{\mu_0 s}\hat{z}$$



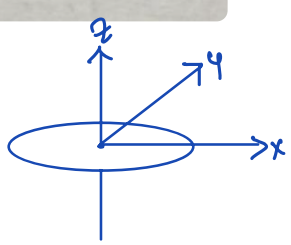
Q6. [3 marks] A thin disk of radius a, with a uniform surface charge density σ, is rotating about its axis (z-axis) with a constant angular frequency ω. There is a uniform magnetic field $\vec{B} = B_0\hat{y}$ perpendicular to the axis of rotation. Determine the torque on the disk.

$$\vec{\tau} = \vec{m}\times\vec{B}$$

$$\vec{m} = \int I\,da = \int \sigma\omega r\,dr\cdot\pi r^2$$

$$i = q/T = \frac{\sigma\times 2\pi r\,dr}{2\pi/\omega} = \sigma\omega r\,dr$$

$$= \sigma\pi\omega r^4/4\,\hat{z}$$



$$\vec{\tau} = \frac{\sigma\pi\omega B_0 R^4}{4}(-\hat{i})$$

Q7. [3 marks]. Two electrons, each possessing a magnetic dipole moment $\vec{m} = m\hat{z}$, are constrained to move along the z-axis. In the absence of any other force, after reaching the equilibrium, how far away will they be from each other?

$$\vec{F}_{dipole} = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{z^2}$$

$$B = \frac{\mu_0}{4\pi r^3}\left[3(\vec{m}\cdot\hat{r})\hat{r} - \vec{m}\right]$$

$$= \frac{\mu_0\left[2m_1\right]}{4\pi r^3}$$

$$F = -\nabla(\vec{m}\cdot\vec{B}) = -\nabla\left(\frac{\mu_0\cdot 2m_1m_2}{4\pi z^3}\right) = \frac{3}{2}\frac{\mu_0 m_1 m_2}{\pi z^4}$$

$$\frac{1}{4\pi\varepsilon_0}\frac{e^2}{z^2} = \frac{3}{2}\frac{\mu_0 m_1 m_2}{\pi z^4} \qquad z = \sqrt{\frac{6\,\mu_0\varepsilon_0 m_1 m_2}{e^2}}$$

$$z = \sqrt{6}\,\frac{m}{ce}$$

INDIAN INSTITUTE OF TECHNOLOGY MADRAS

A

Roll No. | M | t | 2 | 2 | B | 1 | 3 | 6 |

Name : Vansh Jain

Total No. of Pages

Quiz I ☐ Quiz II/ Mid-Sem ☑ End-Semester ☐ Make-up ☐ Date : _____

Semester & Degree : _____ Course No. _____ Part : _____

Question No.	1	2	3	4	5	6	7	8	9	10
Marks										

11	12	13	14	15	16	17	18	19	20	Total

Answer on both sides of the paper including the space below

Q4]

$\vec{B} = B_0 \hat{\phi}$

We know that,

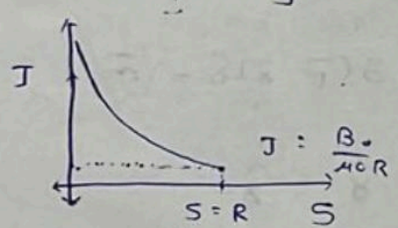
$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}$

Here, $B_\phi = B_0$

$\therefore \left(-\frac{\partial B_0}{\partial z}\right)\hat{s} + \frac{1}{s}\left[\frac{\partial}{\partial s}(sB_0)\right]\hat{z} = \mu_0 \vec{J}$

(2)

$\therefore \vec{J} = \frac{B_0}{\mu_0 s}\hat{z}$ ✓ $\quad \therefore (J)_{s=R} = \frac{B_0}{\mu_0 R}\hat{z}$



✓

0.5

Q5] $\vec{J}_p = \frac{\partial \rho}{\partial t} = ax\hat{i} + by\hat{j}$ ✓

$\vec{I}_p = \int_A \vec{J}_p \cdot d\vec{A} = \int_{x_1}^{x_2} a x\, dx\, dy + \int_{y_1}^{y_2} b y\, dy\, dx$

Which is clearly in exact differential form & hence I_p is conservative

Q6]

$\omega, \vec{B} = B_0 \hat{j}$

We know that,

$\vec{\tau} = \vec{r} \times \vec{B}$

Here $\vec{r} = \int dI \cdot A$

$= \int_0^R \sigma \omega r dr \pi r^2$ ... $\because dI = \sigma v$

$= \frac{\pi \omega \sigma R^4}{4} \hat{k}$ $\because v = \omega r$

Now,

$\vec{\tau} = \frac{\pi \omega \sigma R^4 B_0}{4} (\hat{k} \times \hat{j})$

$= -\frac{\pi \omega \sigma R^4 B_0}{4} \hat{i}$

Q7]

Here, we know

① $m\hat{z}$

② $m\hat{z}$

B (mag dipole) $= \frac{\mu_0}{4\pi r^3} (3(\vec{m} \cdot \hat{r})\hat{r} - \vec{m})$

Here $\vec{m} = m\hat{z}$, $\hat{r} = \hat{z}$

$\therefore \vec{B}(z) = \frac{\mu_0}{4\pi z^3} (3\vec{m} - \vec{m})$

$= \frac{\mu_0}{4\pi} \frac{2m\hat{z}}{z^3}$

Force on 1 due to 2,

$= \nabla (m_1 \cdot B_2)$

$= \nabla \left(\frac{\mu_0 m^2}{2\pi z^3} \right)$

$= -\frac{3\mu_0 m^2}{2\pi z^4} \hat{z}$

Also the force due to electric field is $\frac{1}{4\pi\epsilon_0} \frac{(-e)^2}{z^2} \hat{z}$

$\therefore$ Fnet $= \frac{e^2}{4\pi\epsilon_0 z^2} - \frac{3\mu_0 m^2}{2\pi z^4}$

∴ Equating the force to zero we get,

$z^2 = \frac{G\mu_0 m^2}{e^2}$ $\quad \therefore z = \sqrt{G\mu_0}\frac{m}{e}$

The two electrons need to be at a distance of $\sqrt{G\mu_0}\frac{m}{e}$ where, e is magnitude of charge of e⁻

3

(a) (i)

$\Phi_2 = M_{21} I_1$

$\Phi_2 = \vec{B_1} \cdot \vec{A_2}$

For B_1,

$B_1 = \frac{\mu_0}{4\pi r^3} [3(\vec{m_1} \cdot \hat{r})\hat{r} - \vec{m_1}]$

$\vec{m_1} = (I_1 a_1)\hat{i}$

$\hat{r} = \hat{i}$

$r = (10a)$

$\therefore B_1 = \frac{\mu_0}{4\pi (10a)^3} [2 I_1 a_1]\hat{i} = \frac{\mu_0 I_1 a_1}{2\pi (10a)^3}\hat{i}$

$\vec{A_2} = \frac{a_2}{\sqrt{2}}\hat{i} + \frac{\sqrt{3}a_2}{2}\hat{j}$

$\therefore \vec{B_1} \cdot \vec{A_2} = \frac{\mu_0 I_1 a_1 a_2}{4\pi (10a)^3} = M_{21} I_1$

3

$\therefore M_{21} = M_{12} = \frac{\mu_0 a_1 a_2}{4\pi (10a)^3}$

b) The flux should remain same through coil

2 $\therefore M_{21} = M_{12} = \frac{\mu_0 a_1 a_2}{4\pi (10a)^3}$

$\therefore \Phi_1 = \frac{\mu_0 a_1 a_2}{4\pi (10a)^3} I_2 = L_1 I_1$

$\therefore L_1 = \frac{\mu_0 a_1 a_2 I_2}{4\pi (10a)^3 I_1}$

$W = \frac{1}{2} L_1 I_1^2 = \frac{1}{2} \frac{\mu_0 a_1 a_2 I_1 I_2}{4\pi (10a)^3}$

IIT Madras

Department of Physics

Physics II PH1020

Mid-term examination, 20th February 2021

Summer 2020-21, II Semester — Max mark: 20

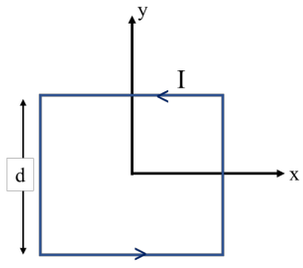
PART-I

Please attempt all 10 questions. (10 x 2=20 marks)

1. The charge distribution in a region is given by $\rho(\vec{r}) = \rho_0 r^a$ where ρ_0 and a are constants of appropriate dimensions.
   (a) Find the expression for electric field $E(\vec{r})$ due to $\rho(\vec{r})$.
   (b) Find the value of a for which $E(\vec{r})$ will increase 16 times if $\vec{r}$ is doubled.

2. The vector potentials
   $\vec{A} = B_0[(bx^2 - z)\hat{j} + y\hat{k}]$ and $\vec{A'} = B_0[-x\hat{i} + (bx^2 - y - z)\hat{j} + y\hat{k}]$
   correspond to the same static magnetic field. Determine the scalar function that relates $\vec{A}$ and $\vec{A'}$.

3. A square loop of sides d, lying on the x-y plane, centered at the origin, carries a steady current I (counter-clockwise when seen from +z side). If an external magnetic field given by $\vec{B} = B_0\hat{i} + B_0 xb\hat{k}$ is applied, find the force and torque on the loop. B_0 and b are constants of appropriate dimensions. Ignore the field due to I.



4. Consider an infinite plane sheet conductor in the x-y plane carrying a uniform surface current density $\vec{K} = K_0\hat{j}$. Calculate the associated magnetic field and magnetic vector potential.

$$\vec{B} = \frac{\mu_0}{4\pi}\int \frac{K_0 da\,\hat{j} \times \hat{\gamma}}{\hat{\gamma}}$$

2. The vector potentials
$\vec{A} = B_0[(bx^2 - z)\hat{j} + y\hat{k}]$ and $\vec{A}' = B_0[-x\hat{i} + (bx^2 - y - z)\hat{j} + y\hat{k}]$
correspond to the same static magnetic field. Determine the scalar function that relates $\vec{A}$ and $\vec{A}'$.

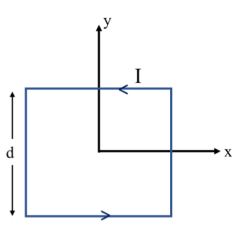
$$\bar{\nabla}\times\bar{A} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 0 & bx^2 - z & y \end{vmatrix} = (1+1)\hat{x} + (2bx)\hat{z} = 2\hat{x} + 2bx\hat{z}$$

$$\bar{\nabla}\times\bar{A_1} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ -x & bx - y - z & y \end{vmatrix} = (1+1)\hat{x} + (2bx)\hat{z} = 2\hat{x} + 2bx\hat{z}$$

$$\nabla\lambda = \bar{A} - \bar{A_1} = B_0[x\hat{i} + y\hat{j}]$$

$$\lambda = \frac{B_0x^2 + B_0y^2}{2} + C$$

3. A square loop of sides d, lying on the x-y plane, centered at the origin, carries a steady current I (counter-clockwise when seen from +z side). If an external magnetic field given by $\vec{B} = B_0\hat{i} + B_0xb\hat{k}$ is applied, find the force and torque on the loop. B_0 and b are constants of appropriate dimensions. Ignore the field due to I.



$$\bar{m} = Id^2\hat{z}$$

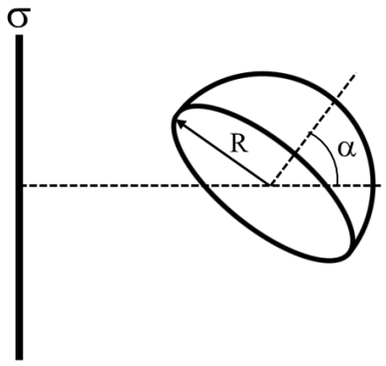
$$\bar{F} = (\bar{m}\cdot\nabla)\bar{B} = md^2\,\frac{d}{dz}(B_0xb) = 0$$

$$\bar{\tau} = \bar{m}\times\bar{B} = Id^2\hat{z}\times(B_0\hat{x}) = B_0Id^2\hat{y}$$

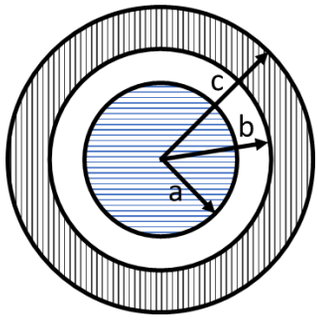
5. Two point charges $-q$ and $+q/2$ are situated at the origin and at $(a, 0, 0)$, respectively, in Cartesian co-ordinates. Find the point on the x-axis where the electric field goes to zero.

6. Two infinite straight-line charges with uniform charge density λ and aligned along the z-axis and separated by distance d . If both the line charges move with a constant velocity $v_0\hat{k}$, for what value of v_0 the net force on the wires is zero?

7. An infinite plane conducting sheet has a uniform surface charge density $+\sigma$. A hemispherical Gaussian surface of radius R has its symmetry axis at an angle α to the normal to the infinite plane, as shown in figure. Calculate the flux of electric field passing through the curved surface of the hemisphere.



8. A coaxial cable has inner core of radius a and outer layer of radii b and c $(b < c)$ as shown in the figure. A uniformly distributed current I runs through the inner core and returns through the outer layer. Calculate magnetic field $\vec{H}(s, \phi, z)$in the region $b \leq s \leq c$.

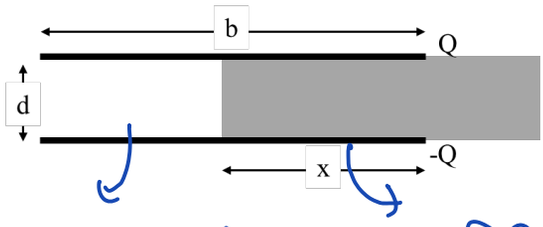


9. An infinitely long cylinder of radius a has its axis along the z-axis. If its magnetization in cylindrical co-ordinate system is given by $\vec{M}(s, \phi, z) = M_0(\frac{s}{a})^2\hat{\phi}$, find the surface bound current density $\vec{K}_b$ and volume bound current density $\vec{J}_b$.

10. A rectangular capacitor of side lengths a and b has separation d between the plates ($d << a, b$). The gap is partially filled with a dielectric with dielectric constant ϵ_r. The overlap distance is x as shown in the figure. The capacitor is isolated and has constant charge Q. Ignore fringing effects.

    (a) Find the capacitance of the capacitor. $C = \frac{\varepsilon_0 ab}{d}$

    (b) Find the energy stored in the capacitor.



$C_1 = \frac{\varepsilon_0 a(b-x)}{d}$ $C_2 = \frac{\varepsilon_0 \varepsilon_r a x}{d}$

Best wishes

$C = \frac{\varepsilon_0 a}{d}[\varepsilon_r x - x + b]$

$E = \frac{Q^2}{2C}$

Some useful equations

Spherical. $d\mathbf{l} = dr\,\hat{\mathbf{r}} + r\,d\theta\,\hat{\boldsymbol{\theta}} + r\sin\theta\,d\phi\,\hat{\boldsymbol{\phi}}; \quad d\tau = r^2\sin\theta\,dr\,d\theta\,d\phi$

Gradient: $$\nabla t = \frac{\partial t}{\partial r}\hat{\mathbf{r}} + \frac{1}{r}\frac{\partial t}{\partial \theta}\hat{\boldsymbol{\theta}} + \frac{1}{r\sin\theta}\frac{\partial t}{\partial \phi}\hat{\boldsymbol{\phi}}$$

Divergence: $$\nabla \cdot \mathbf{v} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 v_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(\sin\theta\, v_\theta) + \frac{1}{r\sin\theta}\frac{\partial v_\phi}{\partial \phi}$$

Curl: $$\nabla \times \mathbf{v} = \frac{1}{r\sin\theta}\left[\frac{\partial}{\partial \theta}(\sin\theta\, v_\phi) - \frac{\partial v_\theta}{\partial \phi}\right]\hat{\mathbf{r}} + \frac{1}{r}\left[\frac{1}{\sin\theta}\frac{\partial v_r}{\partial \phi} - \frac{\partial}{\partial r}(r v_\phi)\right]\hat{\boldsymbol{\theta}} + \frac{1}{r}\left[\frac{\partial}{\partial r}(r v_\theta) - \frac{\partial v_r}{\partial \theta}\right]\hat{\boldsymbol{\phi}}$$

Laplacian: $$\nabla^2 t = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial t}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\frac{\partial t}{\partial \theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 t}{\partial \phi^2}$$

Cylindrical. $d\mathbf{l} = ds\,\hat{\mathbf{s}} + s\,d\phi\,\hat{\boldsymbol{\phi}} + dz\,\hat{\mathbf{z}}; \quad d\tau = s\,ds\,d\phi\,dz$

Gradient: $$\nabla t = \frac{\partial t}{\partial s}\hat{\mathbf{s}} + \frac{1}{s}\frac{\partial t}{\partial \phi}\hat{\boldsymbol{\phi}} + \frac{\partial t}{\partial z}\hat{\mathbf{z}}$$

Divergence: $$\nabla \cdot \mathbf{v} = \frac{1}{s}\frac{\partial}{\partial s}(s v_s) + \frac{1}{s}\frac{\partial v_\phi}{\partial \phi} + \frac{\partial v_z}{\partial z}$$

Curl: $$\nabla \times \mathbf{v} = \left[\frac{1}{s}\frac{\partial v_z}{\partial \phi} - \frac{\partial v_\phi}{\partial z}\right]\hat{\mathbf{s}} + \left[\frac{\partial v_s}{\partial z} - \frac{\partial v_z}{\partial s}\right]\hat{\boldsymbol{\phi}} + \frac{1}{s}\left[\frac{\partial}{\partial s}(s v_\phi) - \frac{\partial v_s}{\partial \phi}\right]\hat{\mathbf{z}}$$

Laplacian: $$\nabla^2 t = \frac{1}{s}\frac{\partial}{\partial s}\left(s\frac{\partial t}{\partial s}\right) + \frac{1}{s^2}\frac{\partial^2 t}{\partial \phi^2} + \frac{\partial^2 t}{\partial z^2}$$

IIT Madras

Department of Physics

Physics II PH1020

Mid-term examination, 20th February 2021

Summer 2020-21, II Semester — Max mark: 10

PART-II

1. Two LIH media has an interface along the x-y plane at $z = 0$. For $z < 0$ (region 1) and $z > 0$ (region 2), the dielectric constants are $\epsilon_{r1} = 3$ and $\epsilon_{r2} = 4$ respectively. A uniform electric field given by $\vec{E} = 5\hat{i} - 2\hat{j} + 4\hat{k}$ exists in region 1.

   (a) Find $\vec{E_2}$. (1 mark)

   (b) Calculate the angles made by $\vec{E_1}$ and $\vec{E_2}$ with the normal to the inter-facial plane (x-y plane) (2 marks)

   (c) Calculate the energy densities in both the regions. (2 marks)

2. A magnetic dipole $\vec{m} = m\hat{k}$ is placed in an otherwise uniform external magnetic field $\vec{B_{ex}} = -B_0\hat{k}$.

   (a) Write an expression for the radial, polar and azimuthal components of magnetic field $\vec{B}(r, \theta, \phi)$ due to $\vec{m}$ only . (1 mark)

   (b) Show that in the combined field there exists a certain spherical surface centered on the dipole through which no field lines pass, Find its radius. (2 marks)

   (c) Sketch the magnetic field lines in the region around the dipole in presence of $\vec{B_{ex}}$. (2 marks)

Best wishes

Department of Physics, IIT Madras

PH1020 Physics II 2018-19 | **Quiz I** | **21.02.2019**

Time: 8:00 am to 8:50 am | *Answer all questions* | **Max. marks: 20**

Name	Roll No.	Old Roll No.(if any)

Instructions: You must write the answers only in the allotted box. All vectors must be indicated clearly using arrows. You may use the reverse side of all pages for rough work. All symbols have their usual meaning unless stated otherwise. All constants are of appropriate dimensions. Further assume that unless otherwise stated all constants are positive. You must use only black or blue ink for writing the answers. Pencil must not be used even in sketches. Calculators, cell phones or any internet connectable device must not be in your possession during the examination.

Exam Hall No.

Total Marks

Q1	Q2	Q3	Q4	Q5	Q6

Use following formula for divergence and curl in spherical polar coordinate

$$\bar{\nabla}\cdot\bar{A}=\frac{1}{r^2}\frac{\partial}{\partial}\left(r^2A_r\right)+\frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}\left(A_\theta\sin\theta\right)+\frac{1}{r\sin\theta}\frac{\partial A_\phi}{\partial\phi}$$

$$\bar{\nabla}\times\bar{A}=\frac{\hat{r}}{r\sin\theta}\left[\frac{\partial}{\partial\theta}\left(A_\phi\sin\theta\right)-\frac{\partial A_\theta}{\partial\phi}\right]+\frac{\hat{\theta}}{r\sin\theta}\left[\frac{\partial A_r}{\partial\phi}-\sin\theta\frac{\partial}{\partial r}\left(rA_\phi\right)\right]+\frac{\hat{\phi}}{r}\left[\frac{\partial}{\partial r}\left(rA_\theta\right)-\frac{\partial A_r}{\partial\theta}\right]$$

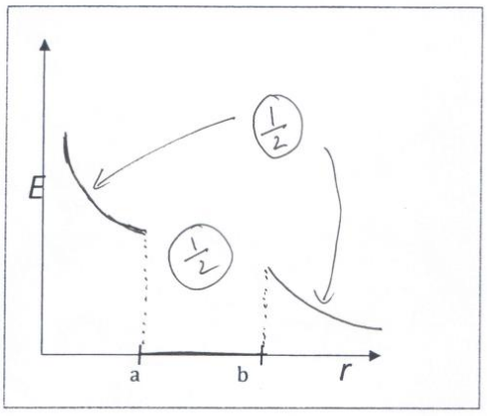
Use following formula for divergence and curl in cylindrical coordinate

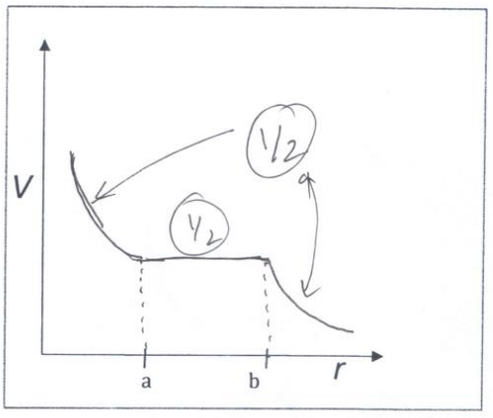
$$\bar{\nabla}\cdot\bar{A}=\frac{1}{\rho}\frac{\partial}{\partial\rho}\left(A_\rho\rho\right)+\frac{1}{\rho}\frac{\partial A_\phi}{\partial\phi}+\frac{\partial A_z}{\partial z}$$

$$\bar{\nabla}\times\bar{A}=\hat{\rho}\left(\frac{1}{\rho}\frac{\partial A_z}{\partial\phi}-\frac{\partial A_\phi}{\partial z}\right)+\hat{\phi}\left(\frac{\partial A_\rho}{\partial z}-\frac{\partial A_z}{\partial\rho}\right)+\hat{z}\left(\frac{1}{\rho}\frac{\partial}{\partial\rho}\left(A_\phi\rho\right)-\frac{1}{\rho}\frac{\partial A_\rho}{\partial\phi}\right)$$

1. In the boxes provided clearly indicate your answers to each of the questions below. The answers to each of these questions **must be a number or an expression or sketch.** Each box carries 1 **mark.**

(a) A positive charge q is placed at the centre of a thick spherical conducting shell of inner radius a and outer radius b. Sketch magnitude of electric field (E) and potential (V) as function of r (distance from the centre). (2 marks)





(b) Polarization within a sphere of radius R is given by $\vec{P} = \frac{P_0}{r}\hat{e}_r$. Where P_0 is a constant of appropriate dimensions (3 marks)

$\sigma_b = \vec{P}\cdot\hat{n} = P_0/R$

$\rho_b = -\vec{\nabla}\cdot\vec{P} = -1/r^2 \frac{d}{dr}(r P_0)$

$= -P_0/r^2$

i. The bound surface charge density on the surface of the sphere is:

P_0/R

ii. The bound volume charge density in the sphere is:

$-P_0/r^2$

iii. The electric field inside the sphere is :

$E_r = -\frac{P_0}{\epsilon_0 r}$ or $\vec{E} = -\frac{P_0}{\epsilon_0 r}\hat{e}_r$

NO VECTOR SIGN THE (1/2)

$\vec{E}\cdot 4\pi r^2 = \frac{\int -P_0/r \times 4\pi r^2 dr}{\epsilon_0}$

$\vec{E} = \frac{-P_0 r}{\epsilon_0 r^2}$

$\vec{E} = -\frac{P_0}{\epsilon_0 r}\hat{r}$

X Indicate the correct alternative(s) in the box provided. No marks will be awarded if any incorrect answer is given. [2 marks]

(A) Work done over a closed path in electrostatics is zero.

(B) A charge kept outside a conducting shell will have its presence felt at the centre of the shell.

(C) Tangential component of electric field is continuous at the interface of two media having surface charge density σ.

(D) For a point charge placed away from the origin, all moments in the multipole expansion except monopole moment identically vanish.

GIVE 1 MARK FOR ONE CORRECT CHOICE
0 FOR ANY WRONG CHOICE

A, C

X In the box provided, indicate whether the following statements are **True** or **False**. [4 × 1 = 4 marks]

(a) The function $\vec{F}(\rho, \varphi, z) = {}^{a}/_{\rho^2}\left(cos\varphi\ \hat{e}_\rho + sin\varphi\hat{e}_\varphi\right)$ represents an electrostatic field. TRUE

(b) Let $\vec{E}_1(r)$ and $\vec{E}_2(r)$ be the electrostatic fields produced by charge distributions $\rho_1(\mathbf{r})$ and $\rho_2(\mathbf{r})$ respectively. Then, the electrostatic energy due to the charge distribution $(\rho_1 + \rho_2)$ is given by $\frac{\epsilon_0}{2}\int(E_1^2 + E_2^2 + 2E_1 \cdot E_2)d\tau$. TRUE

(c) The tangential component of $\vec{D}$ is always continuous across an interface separating two dielectric media which have different dielectric constants. FALSE

(d) For a point charge $+Q$ placed inside a conducting shell, the electrostatic field at any point outside the conducting shell is zero. FALSE

4. An infinitely long thin line charge lies along the z-axis with a uniform charge density λ. An infinitely long annular cylinder of inner radius a and outer radius b with its axis coinciding with z-axis has a volume charge density $\rho_c(\rho, \varphi, z) = k\rho$. Where k is constant of appropriate dimensions. Calculate electric field for all values of $\rho > 0$. (3 Marks)

$\rho < a$

$$E_\rho \, 2\pi\rho L = \frac{\lambda L}{\epsilon_0}$$

$$\vec{E} = \frac{\lambda}{2\pi\epsilon_0 \rho} \hat{e}_\rho \qquad [1]$$

$a \le \rho \le b$ Field due to ρ_c

$$E_\rho \, 2\pi\rho L = \frac{1}{\epsilon_0} \int_{-L/2}^{L/2} \int_0^{2\pi} \int_a^\rho k\rho \; \rho \, d\rho \, d\varphi \, dz$$

$$= \frac{k(\rho^3 - a^3)}{3}\, 2\pi L$$

$$E_\rho = \frac{k(\rho^3 - a^3)}{3\epsilon_0 \rho} \qquad [1\tfrac{1}{2}]$$

Net field $\vec{E} = \left(\frac{\lambda}{2\pi\epsilon_0 \rho} + \frac{k(\rho^3 - a^3)}{3\epsilon_0 \rho} \right) \hat{e}_\rho \; [\tfrac{1}{2}]$

$\rho > b$

$$\vec{E} = \left[\frac{\lambda}{2\pi\epsilon_0 \rho} + \frac{k(b^3 - a^3)}{3\epsilon_0 \rho} \right] \hat{e}_\rho \qquad [1\tfrac{1}{2}]$$

5. A charge configuration in Cartesian coordinates is given as

$$\rho(x') = q\,\delta(x'+a) + q\,\delta(x'-a) - 2q\,\delta(x')$$

Find the monopole, dipole and quadrupole moments of the charge distribution given above. (3 Marks)

$$\text{Monopole} = \int_{-\infty}^{\infty} q\,\delta(x'+a)\,dx + \int_{-\infty}^{\infty} q\,\delta(x'-a)\,dx - \int_{-\infty}^{\infty} 2q\,\delta(x')\,dx$$

$$= q + q - 2q = 0 \longrightarrow 1/2$$

$$\bar{p}_{\text{dipole}} = \int \bar{r}'\,\rho\,d\tau$$

$$= \int_{-\infty}^{\infty} x'\,\hat{e}_x\,q\,\delta(x'+a)\,dx + \int_{-\infty}^{\infty} x'\hat{e}_x\,q\,\delta(x'-a)\,dx + \int_{-\infty}^{\infty} x'\,\hat{e}_x\,(-2q)\,\delta(x')\,dx$$

$$= -qa\,\hat{e}_x + qa\,\hat{e}_x + 0 = 0 \longrightarrow 1/2$$

$$Q_{ij} = \int \rho(\vec{r}')\left[3\vec{r}_i\,\vec{r}_j - \vec{r}^{\,2}\,\delta_{ij}\right]d\tau'$$

$$= 0 \quad \text{for } i \neq j \longrightarrow 1/2$$

$$Q_{xx} = \int \rho(\vec{r}')\left[3x'^2 - (x'^2 + y'^2 + z'^2)\right]d\tau'$$

$$= 2\int x'^2\left[q\,\delta(x'+a) + q\,\delta(x'-a) - 2q\,\delta(x)\right]dx$$

$$= 2\left[qa^2 + qa^2 + 0\right] = \underline{4qa^2} \longrightarrow 1$$

$$Q_{yy} = Q_{zz} = -2q - \frac{Q_{xx}}{2} = \underline{-2qa^2} \longrightarrow \frac{1}{2}$$

X. Charge density in one dimension in a region $0 < x < a$ is given by $\rho = \rho_0 \frac{x}{a}$. Calculate the potential and the electric field in the region $0 < x < a$. The boundary conditions that has to be satisfied are $E = 0$ at $x = 0$ and $V = 0$ at $x = a$. (3 Marks)

$$\nabla^2 \phi = -\frac{\rho_0}{\epsilon_0}\frac{x}{a} \qquad \left(\frac{1}{2}\right)$$

$$\frac{d\phi}{dx} = -\frac{\rho_0 x^2}{2a\epsilon_0} + A = -E_x$$

$E = 0$ at $x = 0 \Rightarrow A = 0$ (1)

$$\frac{d\phi}{dx} = -\frac{\rho_0 x^2}{2a\epsilon_0} = -E_x \qquad \left(\frac{1}{2}\right)$$

$$\phi = -\frac{\rho_0 x^3}{6a\epsilon_0} + B$$

$\phi = 0$ at $x = a \Rightarrow B = \frac{\rho_0 a^3}{6a\epsilon_0}$ (1)

$$\phi = \frac{\rho_0 (a^3 - x^3)}{6a\epsilon_0}$$

1. In the boxes below provided clearly indicate your answers to each of the questions. (Each box carries 1 **mark).**

a) A cylinder of length $2L$ and radius L centred at the origin with its axis oriented along the z-axis has a magnetization $\vec{M} = k\rho^2 \hat{z}$, where k is positive constant of appropriate dimension. Assume there are no free currents.

i. The bound surface current density on the curved surface of the cylinder is

$$\vec{K}_b = kL^2 \hat{e}_\phi$$

ii. The bound volume current density in the cylinder is

$$\vec{J}_b = -2k\rho \hat{e}_\phi$$

iii. Magnetic dipole moment of the cylinder is

$$\vec{m} = \pi k L^5 \hat{e}_z$$

iv. Magnetic field $\vec{B}(r)$ for $r >> 2L$ is

$$\vec{B}(r) = \frac{\mu_0 k L^5}{4r^3}\left[2\cos\theta\,\hat{e}_r + \sin\theta\,\hat{e}_\theta\right]$$

or $\frac{\mu_0 k L^5}{4r^3}\left[3(\hat{e}_r\cdot\hat{e}_z)\hat{e}_r - \hat{e}_z\right]$

b) A thin insulating rod of length L lying on the x-y plane has line charge density $\lambda = k\rho^2$ glued to it. The rod has one end fixed at origin and rotates with angular velocity $\omega_0 \hat{z}$.

i. The current due to infinitesimal line element of the rod is

$$\frac{\omega_0 k \rho^2}{2\pi} d\rho\, \hat{e}_\phi$$

$dq = k\rho^2 ds$

$di = \frac{dq}{T} = \frac{k\rho^2 ds \omega_0}{2\pi}$ e_θ

ii. The magnetic field at the origin is

$$\frac{\mu_0 \omega_0 k L^2}{8\pi}\hat{e}_z$$

$\frac{\mu_0 k R^2 \omega_0}{8\pi}\hat{z}$

$B = \int \frac{\mu_0 di}{2s}$

$= \int \frac{\mu_0 \cdot k\rho^2 d\rho \omega_0}{4\pi s}$

$= 0$

c) Direction of vector potential for a current carrying infinite solenoid with its axis along z-axis is

$$\hat{e}_\phi \text{ or } -\hat{e}_\phi \quad \pm\hat{e}_\theta$$

$\frac{1}{2}$ mark penalty for not indicating direction.

Q.1 (a) (i) $\vec{K}_b = \vec{M} \times \hat{n} = kL^2 \hat{e}_3 \times \hat{e}_\rho = \boxed{kL^2 \hat{e}_\phi}$

(ii) $\vec{J}_b = \vec{\nabla} \times \vec{M} = -e_\phi \frac{\partial M_3}{\partial \rho} = \boxed{-2k\rho \hat{e}_\phi}$

(iii) $\vec{m} = \int \vec{M}\, d\tau = \int_{-L}^{L} \int_{0}^{2\pi} \int_{0}^{L} ke^2 \, e\, d\rho\, d\phi\, dz\, \hat{e}_3$

$= 2\pi \cdot k \cdot 2L \cdot \frac{\rho^4}{4}\Big|_0^L \hat{e}_3 = \boxed{\pi k L^5 \hat{e}_3}$

(iv) $\vec{B} = \frac{\mu_0}{4\pi r^3}\left[3(\vec{m}\cdot\hat{r})\hat{r} - \vec{m}\right]$

$= \frac{\mu_0 m}{4\pi r^3}\left[3\cos\theta\, \hat{e}_r - (\cos\theta\, \hat{e}_r - \sin\theta\, \hat{e}_\theta)\right]$

$= \boxed{\frac{\mu_0 k L^5}{4r^3}(2\cos\theta\, \hat{e}_r + \sin\theta\, \hat{e}_\theta)}$

(b) (i) $dI = \frac{dq}{T} = ke^2 d\rho \cdot \frac{\omega_0}{2\pi} = \boxed{\frac{\omega_0 k e^2 d\rho}{2\pi} \hat{e}_\phi}$

(ii) $\vec{K} = \frac{dI}{dl_\perp} = \frac{\omega_0 k e^2}{2\pi} \hat{e}_\phi$

$\vec{B} = \frac{\mu_0}{4\pi} \int \frac{\vec{K} \times \hat{r}}{r^2} dA = \frac{\mu_0}{4\pi} \frac{\omega_0 k}{2\pi} \int_0^L \int_0^{2\pi} \frac{(\rho^2 e_\phi) \times (-\hat{e}_\rho)}{\rho^2} \rho\, d\rho\, d\Phi$

$= \boxed{\frac{\mu_0 \omega_0 k L^2}{8\pi} \hat{e}_3}$

2. In the box provided, indicate whether the following statements are **True** or **False**.
[4 × 1 = 4 marks]

(a) The equation $\vec{\nabla}.\vec{B} = 0$ implies that magnetic charges do not exist. | TRUE

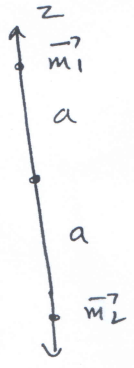
(b) When the magnetic field ($\vec{B}$) crosses a boundary carrying a surface current, the component of magnetic field normal to the surface is discontinuous. | FALSE

(c) In a linear homogenous medium $\vec{J}_b = \chi_m \vec{J}_{free}$. | TRUE

(d) The magnitude of a magnetic field within a long solenoid filled with a diamagnetic materials is greater than that inside an identical solenoid filled with a paramagnetic material. | FALSE

3. Two magnetic dipoles $\vec{m}_1 = m_0 \hat{z}$ and $\vec{m}_2 = -m_0 \hat{z}$ are placed at (0, 0, a) and (0, 0, -a), respectively (Cartesian coordinates). Evaluate the vector potential and the magnetic field (the leading order) on the z-axis for a point z, where $z >> a$. (3 Marks)

$$\vec{A}(r) = \frac{\mu_0}{4\pi} \frac{\vec{m} \times \hat{e}_r}{r^2}$$



$\vec{m} = \pm m_0 \hat{e}_z$; $\hat{e}_r = \hat{z}$, $\vec{A} = 0$

$\vec{A} = 0$ ON z-axis → (1)

$$\vec{B}_{dip} = \frac{\mu_0}{4\pi}\left[\frac{3(\vec{m}\cdot\hat{e}_r)\hat{e}_r - \vec{m}}{r^3}\right] \rightarrow \left(\frac{1}{2}\right)$$

From m_1 $\quad \vec{B}_{dip} = \frac{\mu_0}{4\pi}\left[\frac{2m_0}{(z-a)^3}\right]\hat{e}_z \rightarrow \left(\frac{1}{2}\right)$

m_2 $\quad \vec{B}_{dip} = \frac{\mu_0}{4\pi}\left[\frac{-2m_0}{(z+a)^3}\right]\hat{e}_z$

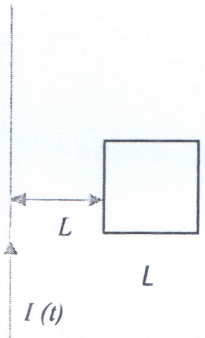
Net field $\quad \vec{B}_{dip} = \frac{\mu_0 m_0}{2\pi}\left[\frac{1}{(z-a)^3} - \frac{1}{(z+a)^3}\right]\hat{e}_z \rightarrow \left(\frac{1}{2}\right)$

$$= \frac{\mu_0 m_0}{2\pi z^3}\left[\frac{1}{(1-\frac{a}{z})^3} - \frac{1}{(1+\frac{a}{z})^3}\right]\hat{e}_z$$

$$\vec{B}_{Net} = \frac{\mu_0 m_0}{2\pi z^3}\left[1 + \frac{3a}{z} - 1 + \frac{3a}{z}\right] = \frac{3\mu_0 m_0 a}{\pi z^4} \rightarrow \left(\frac{1}{2}\right)$$

A square loop of side L and resistance R lies at a distance L from an infinite straight wire that carries time varying current $I(t)$ as follows (see fig.)

$$I(t) = \begin{cases} (1-\alpha t) I_0 & \text{for } 0 \le t \le 1/\alpha \\ 0 & \text{for } t > 1/\alpha \end{cases}$$



a) In what direction does the induced current flow in the loop?
b) Calculate the total charge that passes through any given point in the loop during the time this current flows? (3 Marks)

$$\Phi = \int \vec{B} \cdot d\vec{a} = \frac{\mu_0 I}{2\pi} \int_{-L/2}^{2L} \int_{L} \frac{d\rho}{\rho} dz$$

$$= \frac{\mu_0 I L \ln(2)}{2\pi} \longrightarrow \left(\frac{1}{2}\right)$$

$$\mathcal{E} = I_{loop} R = \frac{dQ}{dt} R = -\frac{d\phi}{dt} = -\frac{\mu_0 L \ln(2)}{2\pi} \frac{dI}{dt} \longrightarrow (1)$$

$$dQ = -\frac{\mu_0 L \ln(2)}{2\pi R} dI$$

$$\therefore Q = \frac{I_0 \mu_0 L \ln(2)}{2\pi R} \qquad (1/2)$$

Current in clockwise — (1)

Assume that the magnetic field of earth arises from a point dipole at its centre and is directed towards the North Pole. Determine the moment of this dipole, given that field at the North Pole is 6×10^{-5} Tesla. Consider the earth as sphere with radius 5×10^{6} meter and $\frac{\mu_0}{4\pi} = 1 \times 10^{-7}$ Newton/Ampere^2. (3 Marks)

$$\vec{B}_{dip} = \frac{\mu_0}{4\pi r^3}\left[3(\vec{m}\cdot\hat{e}_r)\hat{e}_r - \vec{m}\right]$$

$$= \frac{\mu_0 m}{4\pi r^3}\left(2\cos\theta\,\hat{e}_r + \sin\theta\,\hat{e}_\phi\right) \rightarrow \left(\frac{1}{2}\right)$$

$\theta = 0$ $$= \frac{2\mu_0 m}{4\pi r^3}\hat{e}_r \quad - \left(\frac{1}{2}\right)$$

$$\frac{\mu_0}{4\pi}\frac{2m}{r^3} = 6\times10^{-5}$$

$$\frac{10^{-7}\times 2m}{(5\times10^6)^3} = 6\times10^{-5} \quad (1)$$

$$\vec{m} = 3\times10^{-5}\times5^3\times10^{18}\times10^{7}$$

$$= 3.75\times10^{22} \text{ Amp. m}^2 \quad (1)$$

$\hat{e}_2$ or $\hat{e}_r$

Department of Physics
Indian Institute of Technology Madras

Quiz I

Date: Febraury 20, 2018

Time: 08:00 – 08:50 AM

Name: Sampriti Chattopadhyay

Roll No.: CH17B120

Instructor: Santosh.

Batch: 04.

Instructions

1. Please write your name, roll number, instructor name, and batch number.
2. This question paper cum answer sheet booklet contains 7 single-sided pages.
3. The answers have to be written in the boxes provided. Answers written elsewhere in the booklet will not be evaluated.
4. Write the answers, including sketches, with a blue or black pen. Note that answers written with pencils or pens of other colors will not be evaluated.
5. You can use the empty reverse sides for rough work. No extra sheets will be provided.
6. You are not allowed to use a calculator or any other electronic device during the quiz.

For use by examiners

(Do not write in this space)

Page 2	Page 3	Page 4	Page 5	Page 6	Total
5	2	2.	0.5	4	13.5

✦ **Multiple choice questions** (write the correct option in the box provided) [1 x 5 = 5]

1. A spherical conductor of radius a carrying a net charge q is surrounded by a linear dielectric of relative permittivity ϵ_r. The boundary conditions on D at the surface ($r = a$) are

[A] $D^{\perp}_{above} - D^{\perp}_{below} = q, \quad D^{\parallel}_{above} - D^{\parallel}_{below} = 0$

[B] $D^{\perp}_{above} - D^{\perp}_{below} = \frac{q}{4\pi a^2}, \quad D^{\parallel}_{above} - D^{\parallel}_{below} = 0$

[C] $D^{\perp}_{above} - D^{\perp}_{below} = 0, \quad D^{\parallel}_{above} - D^{\parallel}_{below} = \frac{q}{4\pi a^2}$

[D] $D^{\perp}_{above} - D^{\perp}_{below} = 0. \quad D^{\parallel}_{above} - D^{\parallel}_{below} = 0$

B 1

2. A particle of mass m with charge q and initial velocity $\mathbf{v} = v_0\hat{e}_y$ is subject to electric field $\mathbf{E} = E_0\hat{e}_z$ and magnetic field $\mathbf{B} = B_0\hat{e}_x$, where E_0 and B_0 are positive constants. Which of the following is true?

[A] The particle will not undergo deflection only if B = 0.

[B] For E=0, the work done by the B-field transforms into rotational energy of the particle.

[C] The trajectory of the particle will be a helix.

[D] For $\mathbf{v} = \frac{E_0}{B_0}\hat{e}_y$, the trajectory of the particle will be a straightline along $\hat{e}_y$.

D 1

3. Electrostatic potential for a charge configuration is given by $V(x,y) = \frac{a}{\sqrt{\epsilon_0}}(6x + 8y)$ Volts, where a is an appropriate constant and the medium is free-space. The energy density in J/m^3 is

[A] $50a^2$ [B] $100a^2$ [C] $25a^2$ [D] $75a^2$

A 1

4. Consider three particles of unit charge placed at (x, y, z) : (0,0,1), (0,0,0), and (0,0,-1). The z component of the electric field at the point (0,1,1) is

[A] $\frac{1}{4\pi\epsilon_0}\left[1 + \frac{1}{2\sqrt{2}}\right]$ [B] $\frac{1}{4\pi\epsilon_0}\left[1 + \frac{1}{5\sqrt{5}}\right]$ [C] $\frac{1}{4\pi\epsilon_0}\left[1 + \frac{1}{2\sqrt{2}} + \frac{1}{5\sqrt{5}}\right]$ [D] $\frac{1}{4\pi\epsilon_0}\left[\frac{1}{2\sqrt{2}} + \frac{2}{5\sqrt{5}}\right]$

D 1

5. Consider an infinitely long hollow-cylindrical conductor of radius a carrying current I parallel to its axis ($\hat{e}_z$). The magnetic field $\mathbf{B}_{in}$ in the region $r < a$ and $\mathbf{B}_{out}$ in the region $r > a$ are

[A] $\mathbf{B}_{in} = \mathbf{B}_{out} = 0$

[B] $\mathbf{B}_{in} = \frac{\mu_0 I}{2\pi a}\hat{e}_\phi$ and $\mathbf{B}_{out} = \frac{\mu_0 I}{2\pi r}\hat{e}_\phi$

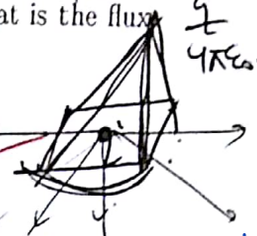
[C] $\mathbf{B}_{in} = 0$ and $\mathbf{B}_{out} = \frac{\mu_0 I}{2\pi r}\hat{e}_\phi$

[D] $\mathbf{B}_{in} = \frac{\mu_0 I}{2\pi a}\hat{e}_\phi$ and $\mathbf{B}_{out} = \frac{\mu_0 I}{2\pi a}\hat{e}_\phi$

C 1

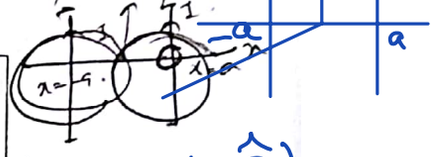
Fill in the blanks (1 mark each, write the answer in the box provided)

6. Consider a pyramid carrying a charge q at the center of its square base. What is the flux through any one face of the pyramid other than the base?



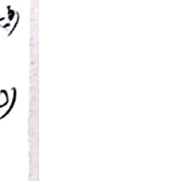
N/A

7. Two infinitely long parallel wires at $x = a$ and $x = -a$ carry current I along $y-$direction. What is $\frac{\partial B_z}{\partial x}$ at the origin?



$\frac{2\mu_0 I}{\pi a^2}$

8. Consider the charge configuration (x, y, z): q at $(0,0,0)$, $-\frac{q}{2}$ at $(1,0,0)$, and $-\frac{q}{2}$ at $(0,1,0)$. The dipolar potential at (1,1,0) is given by

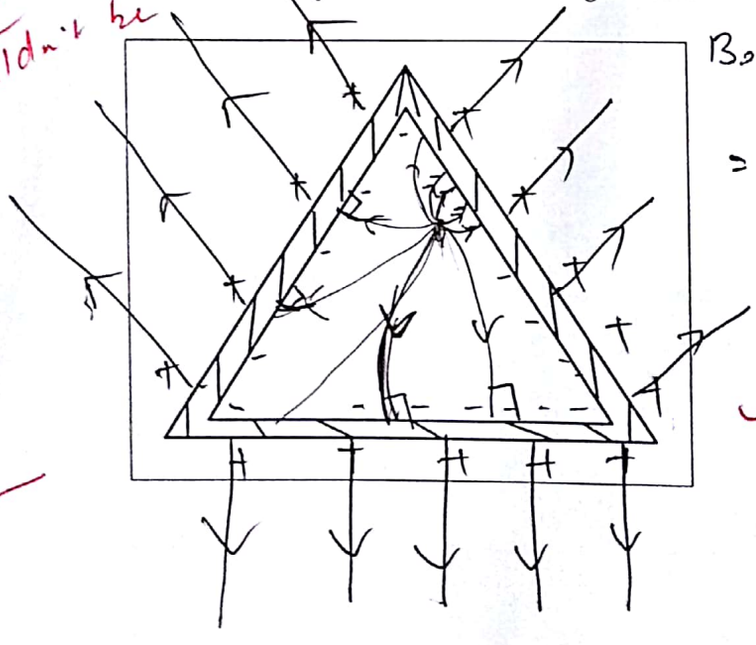


$\frac{-q}{8\sqrt{2}\pi\epsilon_0}$

9. The volume-current density in a straight wire of radius a is $\mathbf{J} = ks\hat{e}_z$, where k is an appropriate constant and s is the distance from the axis. What is the force per unit length on the wire due to a magnetic field $\mathbf{B} = B_0\hat{e}_x$, where B_0 is an appropriate constant?

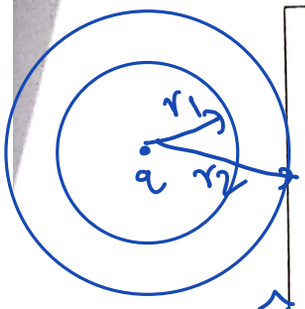
$$-\left(\frac{2\pi k s^3 B_0}{3}\right)\hat{e}_y$$

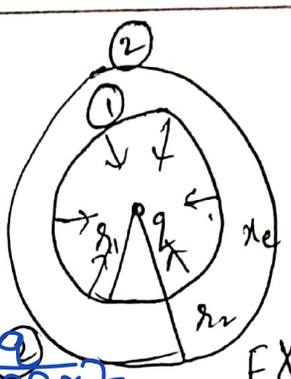
10. A long conducting tube with triangular cross-section encloses a point charge q as shown below. Mark the E-field lines everywhere inside the rectangular box.



✦ Questions with detailed answers (write the calculations and answers within the boxes provid

11. A point charge q is embedded at the center of a spherical shell of linear dielectric material (with susceptibility χ_e, inner and outer radii r_1 and r_2, respectively). Find the electric field everywhere, polarization, and the bound-charge densities ρ_b and σ_b. [1+1+1 marks]





$D\,4\pi r^2 = q$

$D_r = \frac{q}{4\pi r^2}$.

$E\chi_e\varepsilon_0 + E\varepsilon_0 = D$

$\Rightarrow \frac{\chi_e P}{1+\chi_e}\left(\frac{q\hat{r}}{4\pi r^2}\right) = D$.

$E = \frac{D}{\varepsilon_0(1+\chi_e)}$ — (1)

$\Rightarrow \boxed{P = E\chi_e\varepsilon_0 = \frac{D\chi_e}{(1+\chi_e)}}$

For free space.

$D = \varepsilon_0 E$ — (2)

$r < r_1$ $\quad D = \frac{q\hat{r}}{4\pi r^2}$, $P = 0$, from (1) $E = \frac{q}{4\pi\varepsilon_0 r^2}$

$r(r_1, r_2)$ $\quad D = \frac{q\hat{r}}{4\pi r^2}$, $E = \frac{q\hat{r}}{4\pi\varepsilon_0(1+\chi_e)r^2}$, $P = \frac{q\chi_e\hat{r}}{4\pi r^2(1+\chi_e)}$

$r > r_2$ $\quad D = \frac{q\hat{r}}{4\pi r^2}$, $P = E\varepsilon_0\chi_e = \frac{q\chi_e\hat{r}}{4\pi\varepsilon_0 r^2}$

(iii) $E = \frac{q}{4\pi r^2\varepsilon_0}\hat{r}$, $r > r_2$.

$\rho_b = -\nabla\cdot P$

$= -\frac{1}{r^2}\frac{\partial}{\partial r}\left(\frac{q\chi_e r^2}{4\pi r^2(1+\chi_e)}\right)$

$= 0$

$\Rightarrow \boxed{\rho_b = 0}$.

$\sigma_b = -\vec{\nabla}\cdot\vec{P} = 0$

$(\sigma_b)_{inner} = -P(r_1)$

$(\sigma_b)_{outer} = P(r_2)$

σ_b for surface (1) $= P_0(-\hat{r})$

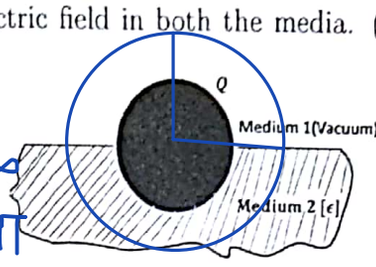
$= \frac{-q\chi_e}{4\pi r^2(1+\chi_e)}$

σ_b for surface (2) $= P\cdot\hat{r}$

$= \frac{q\chi_e}{4\pi r^2(1+\chi_e)}$.

2

12. A conducting sphere of radius R carries charge Q. One half of the sphere is immersed into a linear dielectric with permittivity ϵ, as shown below. The space above this dielectric is vacuum. (a) Find the electric field in both the media. (b) Find the electric displacement



in both the media. (c) Determine the total energy stored in the system. [*Hint: Use the interface between the two media is tangential.*] [1+1+1 marks]

$d\tau = r^2 \sin\theta \, dr \, d\theta \, d\phi$

$r \to 0$ to a, a to ∞

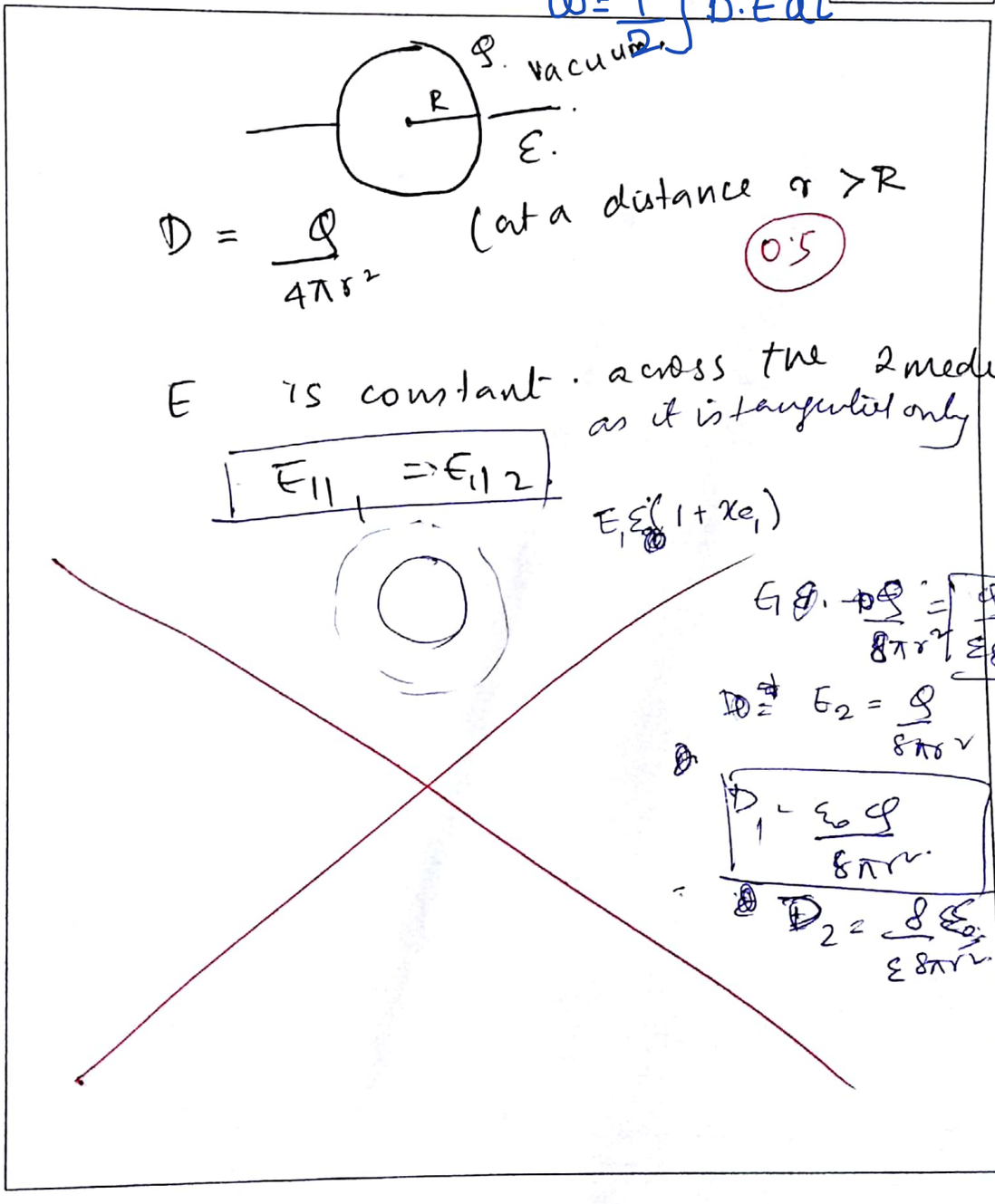
$\theta \to 0$ to $\pi/2$, $\pi/2$ to π

$D = \frac{Q}{4\pi r^2}$ always

outside $E = \frac{Q}{4\pi\epsilon_0 r^2}$

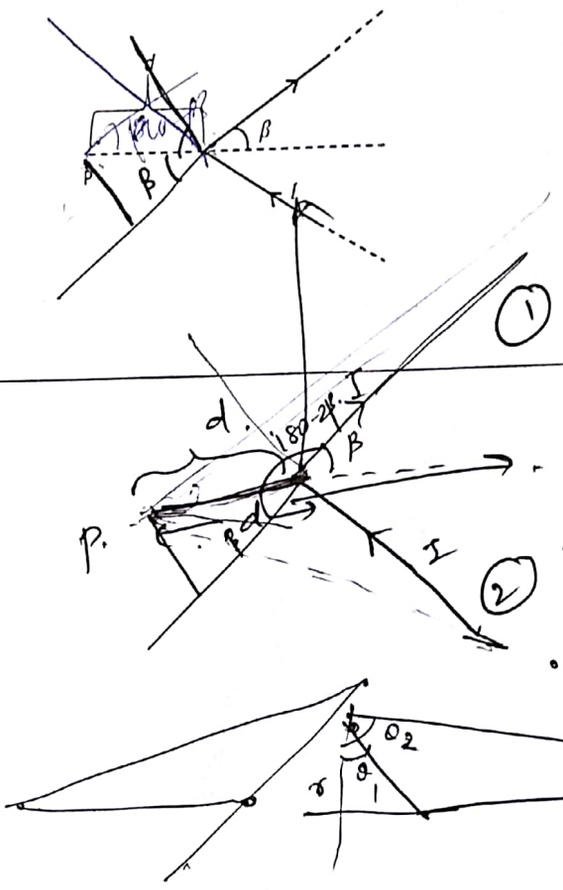
inside $E = \frac{Q}{4\pi\epsilon r^2}$

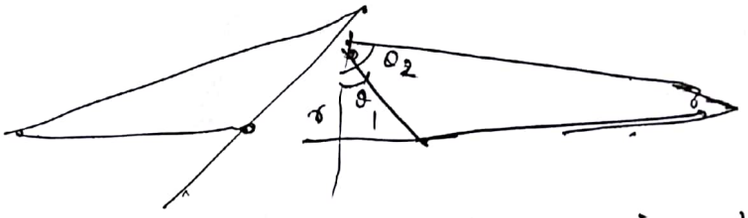
$W = \frac{1}{2}\int \vec{D}\cdot\vec{E}\, d\tau$



13. Consider an infinitely long wire carrying current I. If the wire is bent by an angle 2β as shown in the figure, find the magnetic field at a point P, lying on the angle bisector, at a distance d from the bending point.

4 marks





$$\vec{B} = \frac{\mu_0 I}{4\pi r}\left(\sin\theta_2 - \sin\theta_1\right)$$

$r = d\sin\beta$

$\theta_1 = 90 - \beta$

$\theta_2 = \pi/2$

$$\Rightarrow \frac{\mu_0 I}{4\pi d\sin\beta}\left(\left(\sin\frac{\pi}{2}\right) - \sin(90-\beta)\right)$$

$$\boxed{\vec{B}_1 = \frac{\mu_0 I}{4\pi d\sin\beta}\left(1-\cos\beta\right)\hat{z}} \quad \text{– for wire 1}$$

for the second wire field

$$\vec{B}_{②} = \frac{-\mu_0 I}{4\pi d \sin\beta}\left(-1+\cos\beta\right)\hat{3}.$$

$$= \frac{\mu_0 I}{4\pi d \sin\beta}\left(1-\cos\beta\right)\hat{3}.$$

$$\Rightarrow B_{net} = \frac{\mu_0 I}{2\pi d \sin\beta}\left(1-\cos\beta\right)\hat{3}$$

4

Department of Physics
Indian Institute of Technology Madras

Quiz II

Date: April 3, 2018 Time: 08:00 – 08:50 AM

Name: Sampriti Chattopadhyay **Roll No.:** CH17B120

Instructor: Santosh Nair **Batch:** 3

Instructions

1. Please write your name, roll number, instructor name, and batch number.
2. This question paper cum answer sheet booklet contains **6** single-sided pages.
3. **The answers have to be written in the boxes provided. Answers written elsewhere in the booklet will not be evaluated.**
4. **Write the answers with a blue or black pen. Note that answers written with pencils or pens of other colors will not be evaluated.**
5. **Answers provided without detailed steps for Questions 9 to 11 will not be awarded full marks.**
6. You can use the empty reverse sides for rough work. No extra sheets will be provided.
7. You are not allowed to use a calculator or any other electronic device during the quiz.

Useful expressions

$$\vec{\nabla} \times \vec{v} = \left[\frac{1}{s}\frac{\partial v_z}{\partial \phi} - \frac{\partial v_\phi}{\partial z}\right]\hat{e}_s + \left[\frac{\partial v_s}{\partial z} - \frac{\partial v_z}{\partial s}\right]\hat{e}_\phi + \frac{1}{s}\left[\frac{\partial}{\partial s}(s v_\phi) - \frac{\partial v_s}{\partial \phi}\right]\hat{e}_z$$

$$\vec{\nabla} \times \vec{v} = \frac{1}{r\sin\theta}\left[\frac{\partial}{\partial \theta}(\sin\theta v_\phi) - \frac{\partial v_\theta}{\partial \phi}\right]\hat{e}_r + \frac{1}{r}\left[\frac{1}{\sin\theta}\frac{\partial v_r}{\partial \phi} - \frac{\partial}{\partial r}(r v_\phi)\right]\hat{e}_\theta + \frac{1}{r}\left[\frac{\partial}{\partial r}(r v_\theta) - \frac{\partial v_r}{\partial \theta}\right]\hat{e}_\phi$$

For use by examiners

(Do not write in this space)

Page 2	Page 3	Page 4	Page 5	Page 6	Total
5	$5\frac{1}{2}$	$2\frac{1}{2}+\frac{1}{2}$	3	2	19.5

✦ **Multiple choice questions** (write the correct option in the box provided) [1 x 5 = 5]

1. Mutual inductance of two loops

[A] determines the flux through individual loops due to current passing through themselves.
[B] depends only on the lengths of the loops.
[C] is given by the product of their self-inductances.
[D] depends on the geometries and relative orientation of both the loops.

D

2. A square loop of side a is placed on the x-y plane with one of its corner at the origin and two sides along the positive axes. A time-dependent magnetic field $\vec{B}(t) = ky^3t^2\,\hat{e}_z$, where k is an appropriate constant, is applied. The magnitude of the induced *e.m.f* in the loop is,

[A] zero.
[B] $ka^5t/2$.
[C] ka^4t.
[D] ka^3t^2.

$\mathcal{E} = -\frac{d\phi}{dt} = \int ky^3t^2 a\,dy = \frac{ky^4t^2a}{4}\Big|_0^a = \frac{ka^5t^2}{4}$

$\frac{d}{dt}\left(\frac{ka^5t^2}{4}\right) = \frac{2ka^5t}{4}$

B

3. Which among the following current density distributions corresponds to magnetostatic condition? $\nabla \cdot J =$

[A] $\vec{J}(x,y,z) = 4x\,\hat{e}_x + 5y\,\hat{e}_y - 9z\,\hat{e}_z$
[B] $\vec{J}(x,y,z) = 5y^2\,\hat{e}_y + 10z\,\hat{e}_z$
[C] $\vec{J}(x,y,z) = 2x^2\,\hat{e}_x + 5y^2\,\hat{e}_y + 10z\,\hat{e}_z$
[D] $\vec{J}(x,y,z) = 2x^2\,\hat{e}_x + 10z\,\hat{e}_z$

$\nabla \cdot J = 0$

$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 4x & 5y & -9z \end{vmatrix} = 0$

A

4. The magnetic energy density for a long solenoid with n turns per unit length carrying current I is

[A] $\mu_0(nI)^2/2$
[B] $(\mu_0 nI)^2/2$
C] $(\mu_0 nI)^2/4$
[D] $(\mu_0 nI)^2$

$B^2/2\mu_0 = \frac{\mu_0^2 N^2 i^2}{2\mu_0} \frac{1}{2\mu_0} (\mu_0 n I)^2$

$= \frac{(nI)^2 \mu_0}{2}$

A

5. The magnetic field corresponding to a vector potential, $\vec{A} = -\frac{I_0}{4\pi a^2}\mu_0(x^2+y^2)\,\hat{e}_z$ is

[A] $-\frac{\mu_0 I_0}{2\pi a^2} y\,\hat{e}_x$
[B] $-\frac{\mu_0 I_0}{2\pi a^2}(y\,\hat{e}_x - x\,\hat{e}_y)$
[C] $-\frac{\mu_0 I_0}{2\pi a^2}(x\,\hat{e}_x - y\,\hat{e}_y)$
[D] $-\frac{\mu_0 I_0}{2\pi a^2} x\,\hat{e}_y$

$B = \frac{I_0}{4\pi a^2}\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -2y & 2x & 0 \end{vmatrix}$

$= \frac{I_0}{4\pi a^2}\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & 0 & x^2+y^2 \end{vmatrix}$

B

$\nabla \times A$

$\hat{x}\left(\frac{\partial}{\partial y}(x^2+y^2)\right) - \hat{y}\left(\frac{\partial}{\partial x}(x^2+y^2)\right)$

$\hat{x}\left(\frac{-2yI_0}{4\pi a^2}\right) - \hat{y}\left(\frac{-I_0}{4\pi a^2}2x\right)$

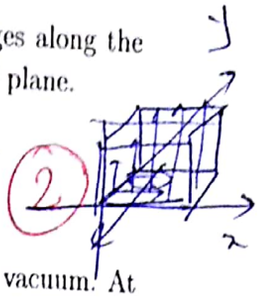
$\frac{-I_0}{4\pi a^2}(x^2+y^2)$

5

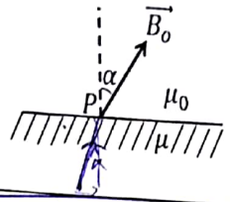
✦ **Fill in the blanks** (2 marks each, write the answer in the box provided)

6. Consider a cube of side a, placed with one corner at the origin and three edges along the positive axes. If the magnetization $\vec{M} = M_0 x\, \hat{e}_z$, find $\vec{J}_b$, and $\vec{K}_b$ on the y = 0 plane.

$J_b = -M_0 \hat{y}$ $\qquad$ $K_b = M_0 x \hat{z}$ (2)



7. Consider the interface of a flat linear magnetic material, of permeability μ, with vacuum. At point P on the interface the magnetic field $\vec{B}_o$ in vacuum makes an angle α with the normal at P. The magnitude of the field in the vicinity of point P inside the material is

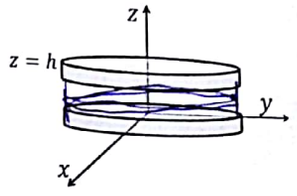


$\frac{B_0 \sin\alpha}{\mu_0} - \frac{B^{\perp}}{\mu} = 0$ $\qquad$ $B^{\perp} = \frac{B_0 \sin\alpha\, \mu}{\mu_0}$

$B_0 \cos\alpha$ $\qquad$ $B_0 \cos\alpha\, \hat{j}$

$B_0 \sqrt{\cos^2\alpha + \frac{\sin^2\alpha\, \mu^2}{\mu_0^2}}$ (2)

8. Consider a parallel-plate capacitor with free space between its circular plates of radius a, negligible thickness, and separation h (where $h << a$). The plates are at $z = 0$ and $z = h$ with z-axis passing through their centers, as shown in the figure. If the voltage applied to the capacitor is $V(t) = V_0 \cos\omega t$, then the maximum value of magnetic field between the plates and direction of $\vec{B}$ are



$B\, 2\pi a = \mu_0 \varepsilon_0 \frac{d\Phi_E}{dt}$

$= \mu_0 \varepsilon_0 \frac{q}{A\varepsilon_0}$

$\vec{B} = \frac{\mu_0 \varepsilon_0 \omega V_0 a}{2h}\, \hat{\phi}$ $\qquad$ alternates between $\hat{\phi}$ to $-\hat{\phi}$. $\qquad$ (1 + 1/2)

$= \frac{\mu_0 \varepsilon_0}{A\varepsilon_0} C \cdot V \cdot A$

$\varepsilon_0 \mu_0 \frac{q}{A\varepsilon_0}$ $\frac{x^2}{t^2}$ $\frac{1}{t}$ $\qquad$ $\frac{1}{c^2}$.

$\frac{\mu_0 \varepsilon_0 A \varepsilon_0 dV}{dt}$ $\qquad$ $\frac{A\varepsilon_0 \cdot d}{B \cdot t}$

$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & 0 & M_0 x \end{vmatrix}$

$(M_0 x\, \hat{e}_3) \times -\hat{y}$

$M_0 x \hat{x}$

$-\hat{y}\,(M_0)$ $\qquad$ (5 1/2) $\qquad$ $\frac{q\Phi}{A\varepsilon_0}$

$B \cdot 2\pi a = \frac{\mu_0 A \varepsilon_0 \omega V_0 \sin\omega t}{h}$

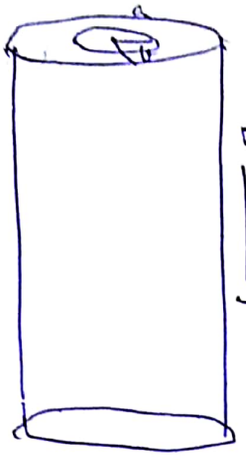
$\frac{A\varepsilon_0}{d}$

$B \cdot 2\pi x = \frac{\mu_0 \Phi \varepsilon_0 \omega V_0 \sin\omega t \times a^2}{h\ 2}$

✦ **Questions with detailed answers** (write the calculations and answers within the boxes provided)

8. A long circular cylindrical shell of inner radius a and outer radius b carries magnetization $\vec{M} = ks^2\,\hat{\phi}$ in the region $a \le s \le b$, where k is an appropriate constant and s is the distance from the axis. Find the magnetic field in the regions $a < s < b$ and $s > b$.

3 marks



$\vec{M} = ks^2\hat{\phi}$

$a < s < b$.

$K_b = ka^2\,\hat{\phi} \times \hat{s}$ at inner surface $= ka^2\hat{z}$

$K_b = kb^2\,\hat{\phi} \times \hat{s}$ at outer surface $= -kb^2\hat{z}$

$J_B = \vec{\nabla} \times \vec{M} = \frac{1}{s}\left[\frac{\partial}{\partial s}(sM_\phi)\right]\hat{e}_z$

$= \frac{1}{s}\left[\frac{\partial}{\partial s} ks^3\right]\hat{e}_s = \frac{3s^2k}{s}\hat{z}$

Ampere's law for $a < s < b$.

$\mu_0 \int J_B \cdot 2\pi s\,ds = 2\pi s \cdot B$

$\mu_0 \int 3s^2 2\pi k = 2\pi s \cdot B$

$\frac{3s^3}{3} 2\pi k \mu_0 = 2\pi s B$

$B = s^2 k \mu_0 \hat{\phi}$

$\mu_0 \left(ka^2 \cdot 2\pi a + \int_a^s 3sk\,2\pi s\,ds\right) = 2\pi s B$

$\Rightarrow \mu_0 2\pi ka^3 + \left[\frac{3k s^3}{3} \cdot 2\pi\right]_a^s = 2\pi s B$

$\Rightarrow \mu_0 \left(2\pi ka^3 + \frac{3k}{3} s^3 2\pi - 2\pi ka^3\right)$

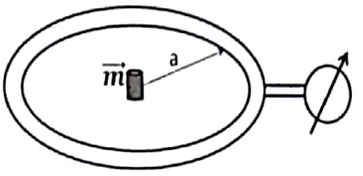
$B = \mu_0 k s^2 \hat{\phi}$ for $a < s < b$

for $s > b$

$\mu_0 \left(2\pi ka^3 + k \cdot 2\pi - 2\pi ka^3 - k2\pi b^3\right)$

$= 2\pi s B$

16. A tiny magnet of magnetic moment $\vec{m}$ is placed at the center of a thin coil of radius a containing N turns. The coil is connected to a ballistic galvanometer. The resistance of the circuit is R. When the magnet was rapidly removed away from the coil a charge q passed through the galvanometer. Find the magnetic moment of the magnet if $\vec{m}$ is perpendicular to the plane of the coil. 3 marks



$\mathcal{E} = L\frac{dI}{dt} - R\,dI = 0.$

$\frac{d\phi}{dt} - L\frac{dI}{dt} - R\frac{dq}{dt} = 0.$

$\Delta\phi - L\Delta I - R\Delta q = 0.$

$\Delta I = 0$

$\Rightarrow \boxed{\Delta\phi = R\Delta q}$

~~Let area of the small magnet be S.~~
The flux will be the same if same current I is passed through the circular coil Bcircular coil at centre

$\Rightarrow N\left(\frac{\mu_0 \cdot I}{2\pi a}\right) \times S = R\Delta q = \frac{\mu_0 I}{2a}.$

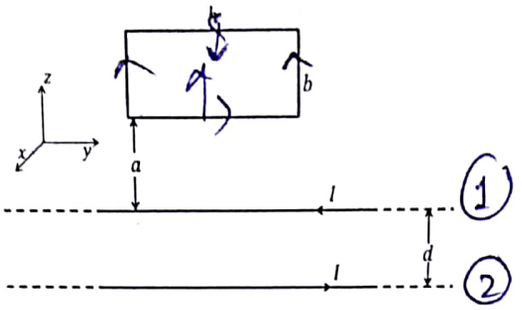
Where S is the area of the small magnet Also $\mu_m = IS$

$\Rightarrow \frac{N\mu_0 \mu_M}{2a} = Rq.$

$\Rightarrow \vec{M} = \left(\frac{Rq2a}{N\mu_0}\right)\hat{k}.$

3

X. Consider two long wires carrying same current $I(t)$ but in opposite directions, as shown in the figure. A rectangular loop is fixed in the plane of the two wires and the relevant dimensions are shown in the figure. If the currents in the wire change at the rate dI/dt then (a) calculate the induced *e.m.f.* in the loop (b) If dI/dt is positive, then what is the direction of force on the loop?



3 marks

$$\phi_1 = \int_a^{a+b} \left(\frac{\mu_0 I \hat{3}}{2\pi s}\right) h\, ds\, \hat{3}$$

$$= -\frac{\mu_0 I h}{2\pi} \ln\left(\frac{b+a}{a}\right)$$

$$\phi_2 = \int_{a+d}^{a+d+b} \left(\frac{\mu_0 I \hat{3}}{2\pi s} h\, ds\, \hat{3}\right)$$

$$= \frac{\mu_0 I}{2\pi} h \ln\left(\frac{a+d+b}{a+d}\right)$$

$$\phi_1 + \phi_2 = \frac{\mu_0 I h}{2\pi}\left(\ln\left(\frac{(a+d+b)}{(a+d)}\right) - \ln\left(\frac{b+a}{a}\right)\right)$$

$$\mathcal{E} = -\frac{d\phi}{dt} = -\mu_0 \cdot \frac{dI}{dt}\frac{h}{2\pi}\left(\ln\frac{(a+d+b)}{(a+d)}\frac{a}{(b+a)}\right)$$

If $\frac{dI}{dt} > 0 \Rightarrow$ ~~flux out~~ Net increase in flux is into the plane

$\Rightarrow$ flux due to current is out of the plane.

$\Rightarrow$ Net force is in upward dir^n $(d\vec{l} \times \vec{B})$